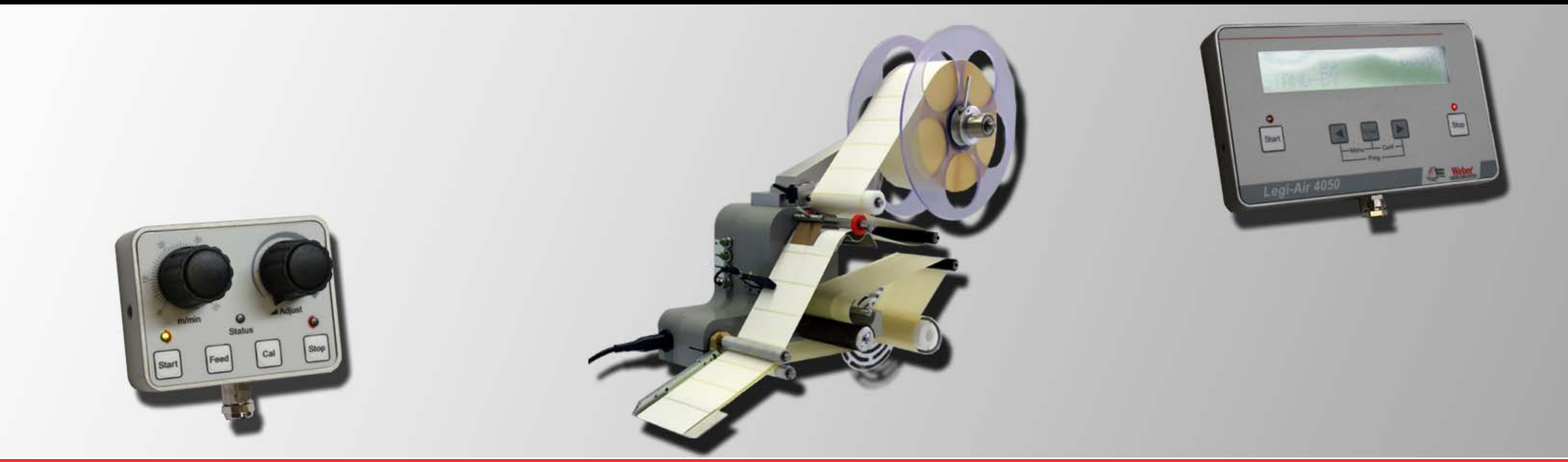
Original Instructions

Manual for Installation, Operation and Maintenance

ALPHA COMPACT

Part number of documentation 32708612



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32708612

Contents

1. General Information

General Survey

Congratulations! You have purchased a high-quality labeling system. Our concern is to make sure that you profit from this system to your entire satisfaction over many years. In order to ensure this, we strongly recommend you to let our experienced specialists perform the installation (hints s. page **34**). Please contact our Service-Hotline (s. page **9**). We are available from Mondays to Fridays 24 hours.

Limitation of Liability

All pieces of information and notes of this manual have been arranged in consideration of applicable standards and regulations, state-of-the-art technology as well as our cognition and experiences over many years.
The manufacturer assumes no liability for damages caused by:

- Non-observance of this manual
- Non-observance of the intended use
- Use of unqualified personnel
- Manipulations at the system
- Use of spare parts that are not approved by the manufacturer

The obligations of the supply contract the General Trading Conditions as well as the Terms of Delivery of the manufacturer and the valid legal regulations at the moment of conclusion of a contract generally apply. Technical changes within the scope of improvement and development are subject to change without notice.

Warranty Clause

The warranty conditions are conform to the valid General Trading Conditions of Bluhm Systeme GmbH at the moment of purchase.

Copyright Protection

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Purpose and Scope of this operation instruction

These operating instructions will help you get to know the system and use it properly.

They contain important instructions for the user on how to use the system safely and correctly. Please comply with this information to avoid hazards, minimize repair costs and outages, and to increase the reliability and service life of the machine.

The operating instructions are for the system identified in the title with the stated type number.

The operating instructions must always be available wherever the system is used. They must be read and used by everyone assigned to work with the system.

Printing mistakes, errors, and changes to maintain the state-of-the-art may occur. Illustrations without protection devices may be presented for the sake of illustration.

Hints for Use of this Manual

Please find in the following a detailed explanation of the notations and representations used in this manual.

- Keys and buttons which you must push appear in squared brackets.

  Example: Push [Enter] - button to save changes...

- Procedures which should be followed in a specific order are listed in numbered paragraphs.

Step	Procedure
1	Disconnect power plug...

- Messages which appear on a controller display are shown in a text box:

```
ALPHA.03a            0000
C=0   004.3   03.6    50.0
```

- Figures and drawings are numbered serially in the particular chapter. For example **„Fig. 2-1“** is the first figure in chapter 2. Components are marked in the figure via numbers. For example: “Fig. 2-1 Pos. 1“ is the first component in chapter 2 in the first figure.

Service Hotline

The technical Service-Hotline is available from Mondays to Fridays 24 hours. In urgent cases spare parts can be dispatched until approx. 22:00 p.m.

Tel :	**+49 (0)2224 - 7708 - 440**
Fax :	**+49 (0)2224 - 7708 - 21**
E-Mail :	**hotline-ed@bluhmsysteme.com**

If failures at the labeling system occur, you should be prepared with the following information when being asked on the phone:

- Detailed error description.
- All information on the name plate of the labeler.
- When did the error occur for the first time?
    - After loading new label material/ribbon?
    - After changes in the setup of the labeler?
- All information about the PLC signals in case of a failure of a print-apply cycle.

Prior to call our hotline service, please have a look at the manual (chapter "Troubleshooting’’) for potential references to eliminate the error.

Our main effort is to keep the service hotline always possibly free for urgent cases.

However we beg your pardon that our hotline also refers to written information provided in this manual.

Explanation of Technical Terms

Technical Term	Explanation
Adhesive Strings	Leaked adhesive at the label edge may adhere the label at the label liner. The printed label adheres to the liner and can thus not be fed to the Tamp.
Air Assist	A stream of air pushing against the bottom of the Tamp/blow box during label feed until label is fixed by vacuum
Air Assist Tube	The air assist tube directs air assist by one or more drillings to the bottom side of the label. It is mounted in the section of the peeler bar and can be adjusted.
Air Blast)	The streams of air effusing from the Tamp drill holes during labeling in order to blow the label by air pressure onto the product (blow box by tube nozzles).
Applying Cycle	An entire working sequence of the labeler from product detection to product labeling. See working sequence.
Application Mode	See explanations: Tamp On, Tamp Blow, Blow On and Wipe On.
Applicator	Description for label applicator respectively mimic that moves the tamp.
Blow-On	A contact free application mode in which the tamp takes the printed label by a stationary vacuum grid and applies with air pressure without moving the tamp.
Configuration Mode	Mode, where parameters for applicator configuration can be changed by display.
Contrast	Difference in the brightness of light and dark sections of a figure / print..
CS	Abbreviation for conveying system (s. a.).
Conveying System	Complete unit consisting of conveyor belt and controller..
Conveyor Belt	Assembly line for product transport.
Cycle	Sequence of a labeling procedure, from product detection to product labeling until the tamp arrives in home position.
Cylinder Home-Position Sensor	Sensor for detection of home position when the tamp is in home position.
Dancer Arm	Arm, tensing the label web via spring tension..
Default	See factory setting
DIP-Switches	DIP is the abbreviation for „Dual In-line Package“. DIP switches are tiny switches with 4 to 8 pins. With these pins the inverter state can be changed (open/closed) by a sharp object (e.g. pencil). The applicator configuration is determined by the switch position.

Technical Term	Explanation
Display	Display providing information on status for the user.
Display-Controller	Labeler-Controller with integrated display for text information (s. Display) and keypad for operation as well as programming**.**
Edge Detection	The edge (front or back) of a product or label that will cause the detector to send a detection signal.
EEPROM	Abbreviation for “Electrically Erasable Programmable Read-Only Memory“. Memory unit on which data can be read and written (programmed). The memorized information will not get lost and remain also without power supply.
Factory Setting	Comprises all basic adjustments of the system after manufacturing which may differ from the condition after a system setup. Software parameters are reset by a reset to the factory setting (default values) and formerly adjusted values are lost.
Factory installable option	Options that are installed and tested in the factory with receipt of system order.
Field installable option	Field installable options are available ex-stock and may by installed by the customer on-site at the labeler.
Home Position	Basic or home position of the tamp at the peeler bar.
HMI	The user interface (HMI) is the place where the user comes into contact with the system. The Alpha Compact can be equipped for selection either with HMI panel or HMI display.
Input-Mode	Mode in which a numeral value in the display can be adjusted higher or lower.
Labeler	Application System for an automatic application of labels.
Label Applicator	See Labeler.
Label Feed	Feeding of a label. Via the [Enter]-button the drive motor of the labeler-rewinder is activated and transport of the label web is started (dancer arm loosens the unwinder brake band). The fed label is peeled off the liner at the peeler bar.
Label Gap Sensor	See Label Sensor.
Label Gap	Gap between two labels on the label web. The gap is identified by a label sensor of the labeler.
Label Liner	Siliconized liner paper that the labels are affixed.
Labeling Mode	Equates to normal mode. The labeler is ready for the application of labels.
Label Out	An optical sensor (reflective light sensor) for identifying the end of the label roll.
Label Sensor	An optical sensor for identifying the label gap.

Label Size	Shows the label format width x length (in feed direction of the label) in mm (millimeter).
Leading Edge	The front edge of the product/label that triggers the label application (see also Edge Detection).
LED	Light Emitting Diode.
Low Label Warning	An optical sensor (reflective light sensor) for identifying a preset minimum of the (adjusted) label roll diameter for a warning.
Mandrel	A driven axis with a device for fixing the label web. The mandrel serves for unwinding the label web
Manometer	See Maintenance Unit
Maintenance Unit (FR-Group)	Unit contains: Air pressure regulator manometer for displaying the air pressure value (in bar) Fast Closing Valve Water separator for draining the possible present perspiration water manually.
Opacity	The transparency of a material can be measured via a light barrier and is called opacity. The quantity of the radiated light is related to the arriving light. The smaller the relation, the smaller the opacity.
Peeler Bar	Metal edge of the applicator for peeling off the label.
Photocell	Product sensor. The sensor identifies the product and activates the trigger signals. See trigger.
Piston	The piston acts on the shaft for the linear movement of the cylinder and transforms air pressure into movement.
PLC	Programmable logic controller.
Potentiometer (Poti)	Depending on the rotational position of the potentiometer, time performance of the tamp movement can be regulated and certain settings of the labeler can be adjusted.
Product Delay)	An adjustable delay from product detection until the applying cycle starts.
Product Detector	A sensor for detection of the product. Mostly optical sensors are used (photocells, light barriers or reflective sensors).
PSI	The American measure of air pressure (Pounds per Square Inch Gauge). 1 bar = 14.7 PSI
Rear Edge	See Trailing Edge.
Reset	Command for resetting software systems on default. Usually activated by a certain key combination..
Rewind and Unwind	See Rewinder / Unwinder.

Technical Term	Explanation
Rewinder	Roll holder (normally for 3 inch cardboard cores) for rewinding the label web. The rewinder winds up the web coming from the print engine. It is triggered by a dancer arm (see dancer arm). Rewinders are powered by a motor.
Stroke	Distance which the retracted Tamp covers towards the product.
Tamp	A unit which takes the label via vacuum and transfers it to the product where it is applied.
Tamp Movement Time	Duration of the Tamp movement (extension and retraction) at labeling cycle.
Tamp Pad	Drilled plate of the Tamp on which the peeled off label is transferred.
Tamp-Blow	Contact free application mode where the Tamp takes the label by vacuum, feeds it to the product and blows it on the product.
Tamp-On	Application mode where the Tamp takes on the label by a vacuum, feeding it and pressing it onto the product.
Trailing Edge	The rear end of a product which is used for activating the application (s.a. edge detection).
Trigger-Signal	Signal of a sensor or a PLC, activating the application cycle.
Variable Stroke Sensor	A proximity sensor mounted on the Tamp for identifying the product. On identifying the product the label is either blown on or peeled off.
VAC	Volts AC (Alternating Current). Also designated as ~.
Vacuum Generator (Venturi)	Unit using air pressure for creating a vacuum.
VDC	Volts DC (Direct Current).
Water Separator	See Maintenance Unit.
Web	Label web consisting of a siliconized liner and adhering Labels.
Wipe-On	Labeling mode where the label is peeled off by a peeler blade and directly transferred onto the product. Label feed and product speed have to be identical.

2. Safety Regulations

Behavior in Case of an Emergency

The operating personnel have to be familiar with the operation and the location of safety, accident notification-, first aid- and rescue devices.

What to do in Case of an Emergency?

- If persons or parts of the body or items are jammed in swing parts of the labeler separate the labeler immediately from air pressure and power supply.
- Initiate immediately all required emergency measures for injured persons. Observe valid safety regulations in any case in order to avoid further damages to persons.
- Call medical attendance for injured persons.
- Eliminate all accident causes.

General Safety Regulations

Safety regulations provide information in written and symbol form in order to warn you against dangers and to instruct you to avoid any damage to persons or to properties.

Safety regulations are started by signal words indicating the level of danger.

Safety regulations may be placed directly at the labeler or in documents about this labeler.

Explanation of Danger Degrees

⚠ DANGER	This symbol indicates a hazardous situation which, if not avoided, will result in death or serious injury. All safety regulations have to be observed to avoid any damage to persons.
⚠ WARNING	This symbol indicates a hazardous situation which, if not avoided, could result in death or serious injury. All safety regulations have to be observed to avoid any damage to persons.
⚠ CAUTION	This symbol indicates a hazardous situation which, if not avoided, may result in minor or moderate injury. All safety regulations have to be observed to avoid any damage to persons.
NOTICE	This symbol indicates a hazardous situation which, if not avoided, may result in damage to properties. All safety regulations have to be observed to avoid any damage to properties.

Intended Use

The working reliability of the labeler is ensured only with intended use.

An Intended Use applies if …

- the labeler is used exclusively for automatic labeling of moved respectively stopped products.
- maintenance is only carried out after stopping the labeler.
- the labeler is used with specified products as well as specified labels. Products and labels have to meet all * documented specifications agreed upon between manufacturer and customer.

  * "documented specifications" are normally laid down in the LSS (Labeling Systems Survey) and handed out to the customer with the confirmation of order.

- the labeler is operating in explosion-proof environments (not intended for explosion-risk areas) !
- the labeler does not come in direct contact with food products.
- the labeler is not operating outdoors.
- the labeler is used with an additional pneumatic shutter at the aperture for the Tamp when operating in a wet environment.
- the labeler has additional air conditioning features in the stainless steel cabinet for use in an aggressive air environment (e.g. salted air).
- the labeler has additional air conditioning features in the stainless steel cabinet for use in a dusty environment with unadjusted particles.
- the labeler is used exclusively for industrial purposes.
- all working conditions and instructions, prescribed in this manual, will be observed.
- failures at the labeler affecting the safety have to be reported and immediately resolved by trained and briefed personnel.
- maintenance is kept and performed correctly.
- the labeler is used exclusively under faultless conditions.
- safety equipment is not by-passed or abrogated.
- arbitrary changes at the machine are omitted.
- the labeler is used or operated by adequate personnel, refer to “Authorized Personnel” (s. page 20). These persons must have read and have to be familiar with the content of the manual.

Handling the labeling system without considering one of these points is not for the intended purpose and can cause serious damages to persons or properties.

Reasonably Forseeable Misuse

Another use as fixed in the „Intended Use“ or even more applies as not intended!

For damages caused by not inteded use:

- the operator bears the complete responsibility,
- the manufacturer assumes no liability.

If you do not use the system according to the regulations, risks may occur!

Not inteded uses are e.g.:

- operation in exlosible atmosphere
- the labeler comes in contact with food …

Retrofitting and Changes at the Labeling System

Unauthorized retrofitting and changes at the system lead to an immediate expiration of liability and warranty covered so far by the manufacturer! This is also valid for interventions and program changes at programmable control systems as well as program changes at control units as far as they are not described in this Manual.
The electromagnetic performance of the system can be affected by amendments or changes of any kind.
Do not arrange any changes or amendments at the systems without consultation and written approval of the manufacturer.

Sources of Danger at Labeler

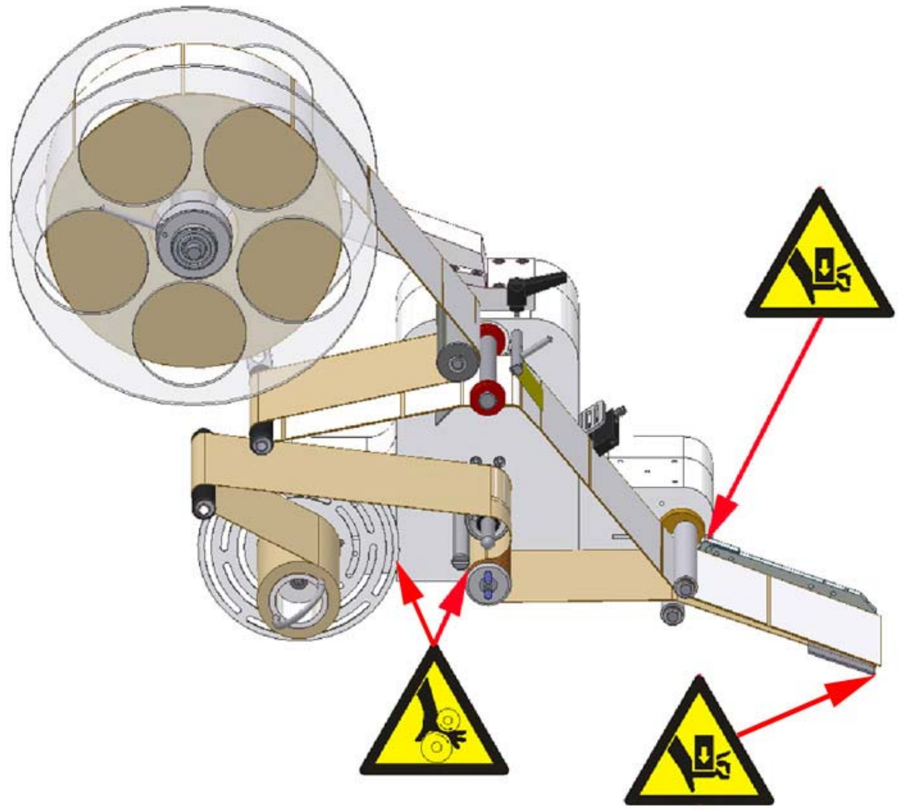


Fig.: 2-1: Sources of Danger at Labeler

The rollers of the unit that brings the objects forward rotate in the opposite direction so that objects may be detected and may be pulled between the rollers. A protective plate here protects the user from injuries. The rewinder is motor-powered. Objects may be detected. There is the danger of being crushed between rewinder table and cabinet. The power is limited to less than 50 N by a round belt and protects the user from injuries. There exists the danger of crushing at the bracket of the peeler blade when the peeler bar swings out. A protective plate protects the user from injuries. In the area of the peeler blade, there may also arise a danger of being crushed by the customer's product feed. Depending on the conveying system and the fed products there may be protective measures by customer necessary.

*²Labeler with optional applicator

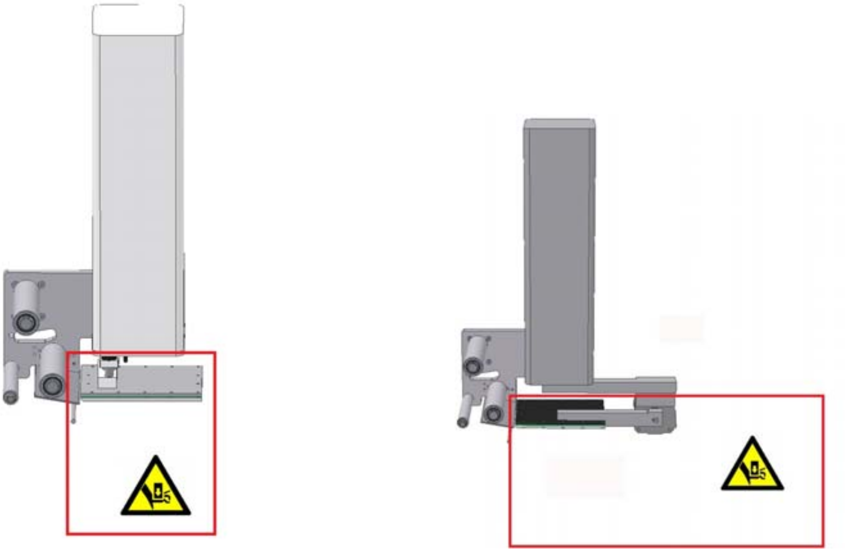


Fig.: 2-2: Sources of danger at the applicator Tamp Blow, rotating tamp and pneumatic peeler blade

All applicators of labelers have been designed according to safety criteria. The arising forces at the Tamp are limited to **50 Newton** for both movements of the stroke, which may be potential risks of injury to persons classified as low (usually reversible). Therefore additional protective devices (e.g. safety guards, protective cabinet…) are not required. The limitation of the Tamp force to 50 Newton is realized by using a smaller cylinder diameter. The prescribed pressure of the compressed air may not be exceeded (see p.25).

*² Only if labeler has the appropriate feature.

Safety Instructions

⚠ WARNING

Danger caused by direct or indirect contact with live parts.

DANGER TO LIFE!
Contact of persons with live parts.

- Before performing any work on electrical equipment of the labeler, separate from power source.
- Work on electrical equipment of the labeler may only be carried out by electricians.
- The electrical equipment of the labeler has to be checked regularly. Damages have to be repaired professionally and immediately.
- Incorrect and faulty connected control units can cause failures/damages and lack of safety.
- To avoid static electricity, ground the labeling system.

⚠ WARNING

Danger caused by highly combustible label material.

FIRE DANGER!

Label web is highly combustible. Risk of injury by fire and smoke.

- Keep clear from naked flame and source of ignition.

⚠ CAUTION

Danger caused by actively triggered movements.

DANGER OF DRAWING IN!
Rotating elements at the labeler, rewinder, web brake and rollers of the friction unit.

- Do not grip in, at or between the moving parts.

⚠ CAUTION

Danger from moved products.

DANGER OF BEING CRUSHED!

At *²pivoted peeler blades the passing products push away the peeler blade. Thus the distance between deflection roller and bracket of peeler blade is decreased. Objects may then be crushed.

At *²fixed mounted peeler blade, objects may be crushed between product and peeler bar.

When using an *²applicator, objects may be crushed during extension and retraction movement or swinging movement between tamp or peeler bar.

- **Maintain a distance from *²peeler blade, bracket of peeler blade, *²tamp pad and the products.**

CAUTION

Health hazard caused by rough surfaces

HEALTH HAZARD!

The friction rollers have a rough surface. Long contact may cause abrasions.

– Do not grip in, at or between moving parts.

CAUTION

Health Hazard due to inappropriate handling of lubricants and detergents

HEALTH HAZARD!

For used lubricants and detergents, the valid information of the safety data sheets of the manufacturers and the valid safety- and disposal regulations have to be observed and followed for each product.

CAUTION

Risk of injury at angles and edges

RISK OF INJURIES!

Sharp angles and edges may cause grazing and gashes at the skin. Keep the working place always clean. The label web builds sharp angles.

– If work is performed close to sharp angles and edges, proceed cautious. Remove parts that are not required.
– In case of doubt wear safety gloves.
– Attention when inserting and changing label web.

NOTICE

Engine (motor) and controller have to be grounded (grounding screw next to motor plug). Short time overloads influence the internal direct-current voltage in a negative way (chassis on 0V potential) and should be avoided.

Remaining Risks

The labeler is constructed for a safe operation. Hazards that are not preventable due to construction purposes are limited as far as possible by protection devices. A certain amount of risk is always existent! The knowledge about the remaining risks assists you to arrange your work safer and to avoid incidents. In order to avoid the dangers, please observe additionally the particular security advice in the single chapters.

Warnings on the labeler

Special hazards arising from the labeler are identified with yellow stickers. The pictograms indicate hazards:

	Danger
	Life-threatening hazard by electrical power
	Crushing hazard
 	Entanglement hazard
	Danger due to hot surface
	Danger due to strong light radiation
	Observe manual

Protection device

Machine cladding

The fixed and screwed machine cladding (cover) protects the user from mechanical and electrical hazards.

Authorized Personnel

Work at the labeler should only be performed by reliable personnel. Please comply with the legal age!

Only trained personnel are allowed to operate the labeler. Trainees, apprentices etc. must be supervised by an experienced person while working at the labeler.

Prior to start running the labeler the operator has to ensure that the manual of the labeler is available to all users of the machine and that the users have read and understood the manual. Only then the system may be put in operation.

The responsibility for the different tasks at the labeler must be clearly specified and kept. There must be no ambiguous authorities for this may put the safety of the users at risk. Arrange a detailed work schedule if several persons work on the machine.

All work on the electrical equipment must be carried out by skilled electricians only.
Failures may be eliminated by authorized personnel only.
All work associated with the assembly, adjustment and maintenance at the machine may be carried out only by trained or instructed personnel.
The operator of the machine must ensure that the personnel are trained in dealing with the integrated control system prior to fix machine errors or maintain the system.

Working Places Operator Personnel

The labeler is an automatic working apply system and does not require any operation for the labeling procedure. For the loading of labels the operation by one person is provided and it is only permitted if the labeler is stopped. The operator side is generally the system front side (see Fig. 4-3).

Personal Protective Equipment

Wear following protective equipment when performing work at the labeler:

SAFETY SHOES

Wear for protection against falling off parts and slipping.

PROTECTIVE CLOTHING

Are tight-fitting clothes with low tensile strength, with tight sleeve and without distant parts.

- Wear a hairnet if applicable.
- Do not wear jewelry or wrist watches.

PROTECTIVE GOGGLES

For protection against splashs of detergents and flying parts.

SAFETY GLOVES

For protection against sharp-edged items.

Personal Protective Equipment for the following tasks.	**Protective Clothing**	**Safety Shoes**	**Safety Gloves**	**Protective Goggles**
System Transport.	X	X	X	
Setting up and connecting of the system.	X	X		
Maintenance Work	X	X		X
Labeling Mode	X	X		
	The documentation of the manufacturer of the single system components has to be observed!			

Disposal

This labeler complies with the RoHS EU-Regulation 2002/95/EG with observance of the fixed using prohibitions and avoiding pollutants.

3. Technical Specifications

Dimensions Labeler (H x W x D in mm)	715 x 310 x 590 (dependent on applicator version)
Weight	approx. 22 kg (without label roll)
Power Connection:	100-240V / 47-63Hz (1~) with protective conductor
Power Consumption:	*“200 W or *² 300°W see type label
Environmental Temperature:	10-38 °C
Environmental Conditions:	20-90 % relative air humidity (non-condensing)
Protection Rating:	IP 20
***² Air Pressure Connection:**	5 – 6 bar (according to **DIN ISO 8573-1)**
	Air Pressure quality according to DIN EN ISO 8573-1 **- Residual Oil Content (Quality Class 5) =** 25 mg/m³ **- Residual Dust (Quality Class 3)=** 5 µm ; 5 mg/m³ **- Residual Water (Quality class 4)=** 6 g/m³ ;+3 C
Maximum Air Pressure Consumption	Up to 200 l/min (under extreme conditions) (0,3 – 3 Liter per applying cycle)

The system needs a noise pulses-free power supply voltage as usual for systems of DP-category. Please pay attention to the recommended quality of power supply voltage. (Ask for Brochure: **Preparations for installation of an electronic print and apply system** of Weber Marking Systems).

*² Only if labeler has the appropriate feature.

Information on Further Dimensions:

Minimum Label Width	10 mm
Maximum Label Width	Depending on version either 120 mm or 150 mm
Minimum Label Length	20 mm
Maximum Label Length	300 mm
Applying Distance (for Tamp-Blow, Blow-Tamp-On)	0-20 mm distance between product and tamp. Dependent on application accuracy and label size.
Label Roll, Outside Diameter:	Ø =**300** mm (corresponds to ca. 450 running meters) core = 76 mm (3 “), outside winding preferred.

Performance Data

Application Accuracy (Wipe-On and Tamp-On-Applicators)	+/- 0,8 mm
Application Rate Wipe-On / Blow Box	Up to 600 labels per min.
Application Rate Tamp-On Applicators	Up to120 labels per min.
Application Performance	Max. 50 m per minute

Signal Connections

Start-Connection (Trigger)	Signaltyp: PNP(Hilfsspannung +24 Volt +/- 7 %)
Stop-Light Barrier	Type NPN (forked light barrier)

Information on Operation

Panel Controller	2 rotary and 4 push buttons
Display Controller	Display: LCD with background lighting, 2-lines 16-digit and 5 push buttons

Noise Level

The measuring in a distance of 1 meter to system surface and 1,60 meter above the floor provided the following noise levels:

(Measuring without optional blowing up-system)

A- Sound Pressure Level	<70 dB (A)

4. Description of the Labeler

Application Field of the Labeler

The ALPHA Compact is used for the automatically labeling of products. Depending on **machine attitude** the labeling can be arranged on the sides or on the top face or the bottom side onto the product without any interruption. Depending on the application a **lefthand (LH)** or **righthand (RH)** version of the labeler is used.

Usually the products are transferred to the labeler by a conveying system.

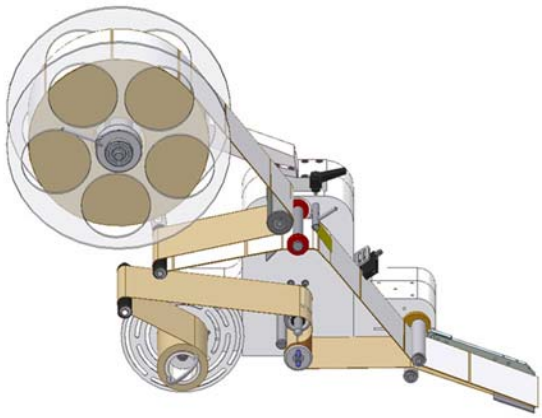


Fig.: 4-1: Alpha Compact Lefthand Version **(LH)**

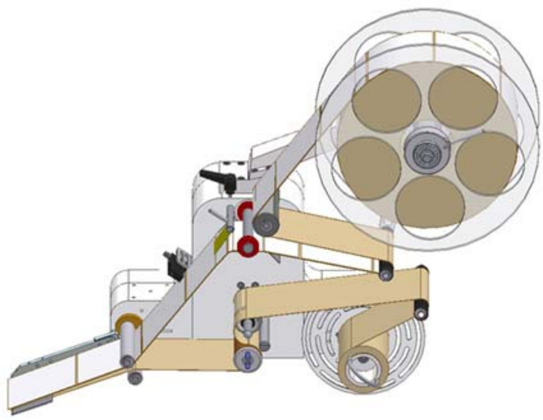


Fig.: 4-2: Alpha Compact Righthand Version **(RH)**

Function of the Labeler

The labels are unwound together with the label liner from a roll by a motor powered pair of draw rollers and pulled around a peel bar (peeler blade). Thereby the labels are peeled off the label liner and stuck on the passing product. The rest of the label liner is motor-driven rewound. If the label material is consumed, the labeler will stop. To activate the labeler again, the label rest-roll and the label liner have to be removed manually and new label material has to be loaded in the labeler.

This kind of label-application is called **Wipe-On** and is the standard applying mode

Application Modi

The Alpha Compact applies the labels onto stopped or moved products. Therefore several application procedures (modi) are possible and can be configured by the HMI display. In application Modi, Tamp-Blow-Mode, Tamp-On-Mode and Blow-On-Mode, optional applicators have to be installed at the Alpha Compact.

Wipe-On

The label is peeled off from peeler blade and applied onto the passing product. This mode is the standard application mode of the Alpha Compact and it is possible without further mounting parts (applicators).

Tamp-Blow-Mode

The label is positioned by an optional applicator to a vacuum tamp and at triggering it is extended to the product and it is then blown off onto the product.

The Tamp-Blow-Mode is often used as the products need to be labeled contactless and it thus presents a low-wear working method.

Tamp-On-Mode

In difference to Tamp-Blow-Mode, the product is touched during labeling. The label is pushed onto the stopped product by the vacuum tamp. This mode requires a change of the applicator configuration and resp. an adjustment of the labeling position.

Blow-On-Mode

The label is forwarded by an optional applicator with tamp by the vacuum and blown off onto the product without any movement of the stroke. The application mode cannot be used in all cases. BLUHM Systems therefore offers a testing under original conditions.

Complete Overview of the Alpha Compact

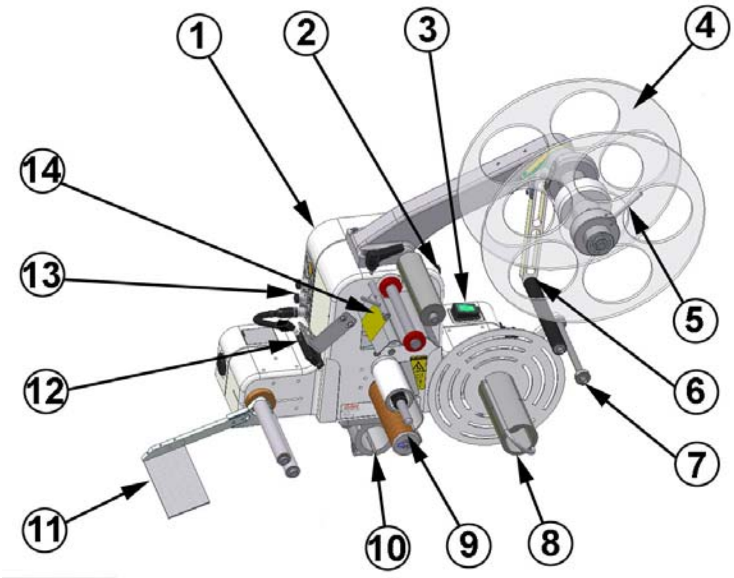


Fig.: 4-3 Alpha Compact Assembly Groups

No.	Description
1	CABINET WITH CONTROLLER
2	CONNECTOR
3	*² ON / OFF SWITCH (OPTION)
4	UNWINDER
5	UNWINDER DISC WITH CLAMPING
6	DANCER ARM FOR UNWINDING
7	DANCER ARM FOR REWINDING
8	REWINDER
9	FEEDING UNIT (FRICTION ROLLERS)
10	CLAMPING FLANGE
11	PEELER BAR (PEELER BLADE)
12	LABEL SENSOR
13	CONNECTORS
14	BELT BRAKE

Subdivision of the product range of the Alpha Compact

See chapter [System Options] p.114.

5. Transport

Delivery

If the labeler is delivered by a carrier, check the packaging for damages. If anything is peculiar, we kindly ask you to claim at the carrier and mark it on the delivery note.

Scope of Delivery

The scope of delivery of the Alpha Compact depends on the ordered options and the customer's application. Please control the scope of delivery when receiving the systems on the basis of the delivery note.

Transport and Unpackaging

Safety Instructions

WARNING

Danger due to lifted weight.

Crashing weights may result in severe injuries or even in death!

- Do not move beneath the lifted weight. The weight may not be banked.
- The position of the center of gravity has to be observed during transport of the system.

WARNING

Danger due to tensioned straps.

Straps are tensioned and may lash back uncontrolled when it is cut through and may cause severe injuries!

- Wear protective goggles and protective gloves.
- Stand laterally outside of the dangerous area.

CAUTION

Danger due to falling parts.

- Wear safety shoes.

NOTICE

To avoid damaging the labeling during transport, do not overturn the labeler, avoid strong vibrations, protect it from humidity and rain. If it is supplied in cardboard boxes, please store at maximum three systems on top of each other. If it is supplied in a wooden box, do not put the wooden box on top of each other.
Remove the packaging material and the transport securing devices only at the site of use, and transport the labeler in its original packaging to the labeling site. If the labeler is not secured, it can tip over easily when transported.

Requirements

- The labeler is unpacked in delivery condition (differences are possible), e.g.:
  - It stands on a pallet
  - It is stretched in.
  - (only in case of protective cabinet) it is secured with additional straps and resp. screwed together to the pallet with its rubber feet.

Required Resources

- Suitable means of transport (fork lift or pallet truck with a payload of at least 600 kg)
- When using a fork lift, drive slowly.
- (Only with protective cabinet) For unloading the system from pallet, a double hand fork lift truck or pallet jack or…
- 2 persons, who have to wear safety shoes.
- Stable support for the labeler.
- (Safety-) steel band scissors for removal of the straps.
- Screw wrench for transportation safety device.

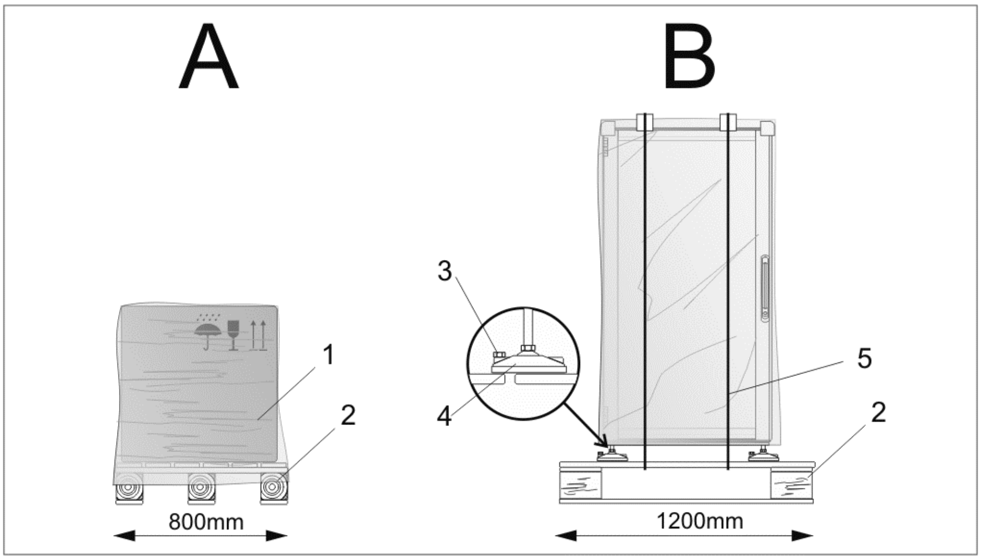


Fig. 5-1: Packaging Examples

Pos.	Description
A	PACKAGING EXAMPLE WITH CARDBOARD BOX OR BOX
B	PACKAGING EXAMPLE WITH PROTECTIVE CABINET
1	STRETCH FOIL
2	PALLET
3	TRANSPORT SECURING SCREW
4	RUBBER FEET
5	STRAPS

Instruction

Please transport the labeler to the installation site as follows:

Step	Procedure
1	Transport the labeler to its installation site (within a radius of 3m). The exact positioning will be arranged during installation by a technician of the Bluhm Weber Group.
⚠ WARNING	Straps are tensioned and may lash back uncontrolled when it is cut through and may cause severe injuries.
2	Remove foil packaging and straps (if available). Without protective cabinet, go on with **Step 5.**
3	Remove all transportation safety device screws, if available, from rubber feet (s. **Fig. 5-2:**).

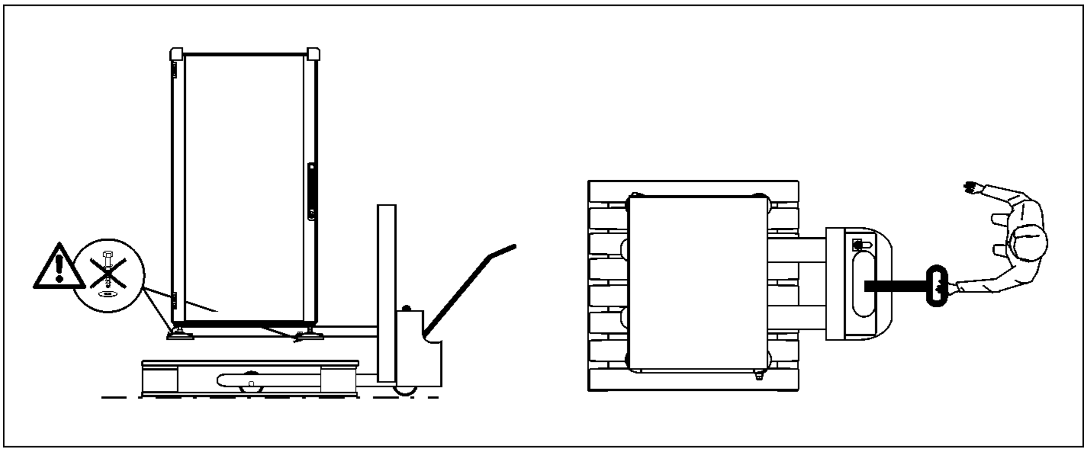


Fig. 5-2: Alpha Compact with Protective Cabinet from Pallet.

4	Lift the labeler with a double hand fork lift truck resp. fork lift truck in shown way from the pallet (s.**Fig. 5-2:**). **OR** Move the labeler from pallet piece by piece diagonally by three persons.Go on with **Step 7.**

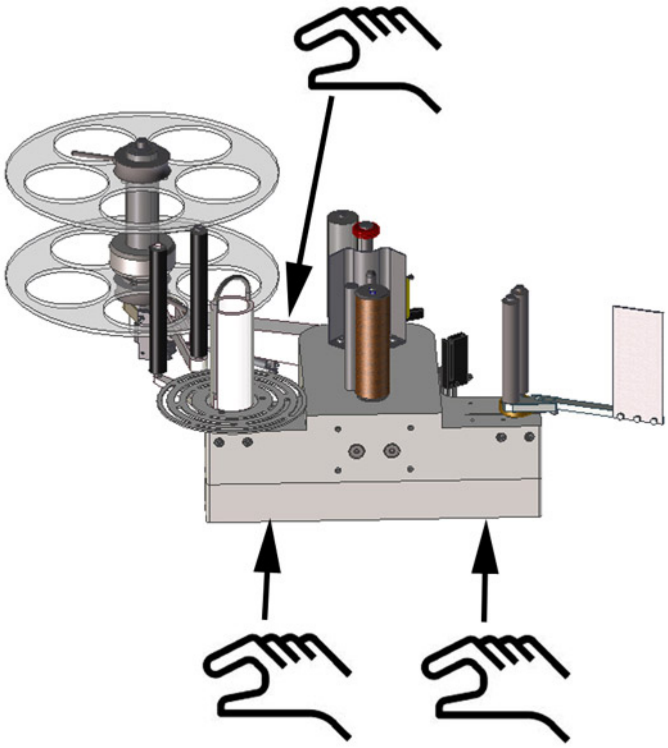


Fig. 5-3: Lifting Alpha Compact

5	Open the carton and remove the packing material and enclosed material.
6	Lift the system with at least two persons at the base plate and the rewind-er (cp. **Fig. 5-3**) out of the packaging and move it with the backside to a stable support. Systems with optional stand mounting can be touched there additionally.
7	Remove all transportation safety devices (marked red tie wraps) to install the system.

Storage Conditions

The environmental conditions for storage of the labeler relate to them of the normal operation. Details see chapter "Technical Data".

Instruction

Please store the labeler safety as follows:

Step	Procedure
1	Label material has to be removed from system.
2	Fix the labeler on pallet(s) and transport the system to its storage place, consider for protection and transport the remarks from above-mentioned section "Transport and Unpackaging".
3	For dust protection, cover the labeler with a cotton cloth or a paper towel. In order to avoid condensate formation, foils may not be used.
4	Before restarting the system, a check of the labeler is required.

6. Installation and Initial Operation

Installation

Only an optimally aligned installation of the system can ensure a continuous operation with a low rate of failures and a minimum wear. For an optimized installation of the system, fine tunings adapted to environmental conditions are essential. For the fine tunings, a complex expert knowledge is required basing on experience with labeling technique.

The complexity of a wear-optimized installation requires a high measure of specialized knowledge and experience, which cannot be obtained completely by reading this manual.

Therefore the installation of the labeler must be made by a technician from the **Bluhm**

Weber Group or examined by a final inspection. Damage or damages based on an incorrect installation, represent no case of warranty.

Furthermore additional information on initial operation and maintenance is available in the Service Manual.

Requirements to the Site of Installation

- Closed and clean room.
- Level surface, solid floor, unevennesses must not exceed 5 mm if stands of the Bluhm Weber Group are used.
- Floor with adequate carrying capacity: 1500 kg/m².
- Low vibration environment.
- Adequate illuminating: 500 Lx.
- No direct irradiation from sun light or heaters.
- The machine must not operate in areas with electrostatic or magnetic fields. This can result in malfunctions of the controller.
- Correct air and power supply according to the chapter "Technical Data".

Placing the Labeler

- The labeler may not contact the product.
- The installation position has to provide sufficient access for user and service technician. In particular at all times, the mains switch / plug must be freely accessible to disconnect the power supply
- Observe that all mounting parts are fixed sufficiently.
- Consider all points of the "Intended Use" in the chapter safety regulations.

*² Only if system is equipped accordingly.

Positioning the Labeler

NOTICE

Property damage caused by improper installation.

Use only the existing factory mounting points.

– Do not drill new mounting points into the cabinet.

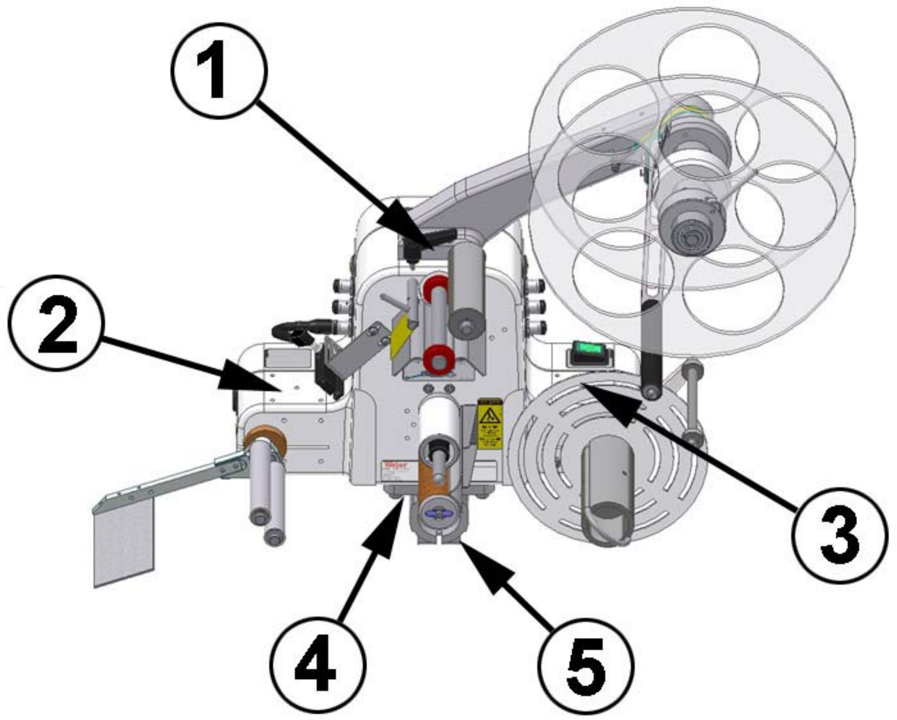


Fig.: 6-1 Mounting Points

No.	Description
1	MOUNTING POINT TOP SIDE
2	MOUNTING POINT LEFT SIDE
3	MOUNTING POINT RIGHT SIDE
4	MOUNTING POINT BOTTOM SIDE

The Alpha Compact is equipped with four possible fixing positions at the cabinet. Depending on the order there are four M8 or four M6 threads at one of the positions. The other 3 mounting points are marked.

The mounting points may not be drilled later. The components of Alpha Compact would be damaged thereby.

When fixing all four threads have to be screwed to a sufficiently stable mounting. The screws have to be at least 8 mm deeply into the thread. Suitable are screws of the property class 8.8 with a tightening torque of 18Nm at M8 and 10Nm at M6. At standard systems at the bottom side a clamping flange is fixed that enables the installation at 40 mm tubes. Many suitable mountings are available in our product range and can be ordered at Weber Marking Systems GmbH.

*² System for fixing at stands of the Bluhm Weber Group!

If the system is fixed at a Stand of the Bluhm Weber Group, you will please follow the installation instructions.

Requirements

- The labeler is completely mounted (in protective cabinet or at stand > s. advice above).
- The labeler is available unpacked and prepared at the installation site (s. chapter "Transport")
- Connections for air pressure and power according to "Technical Data" are installed close to the labeling site (max. 1,50 m away).
- Footprint is fixed, horizontal and flat.

*² Only if system is equipped accordingly.

Required Resources

- Screw wrench (for adjustment of rubber feet)
- Air lever

Instruction

Please position the labeler as follows

Step	Procedure
1	If possible, adjust the labeler at the stand to the lowest position, so the balance point is very low. Pay attention to asperities or changing state of floor during transportation. It might cause an abrupt stop of a roll, so the stand and system could fall.
2	Roll the labeler to the labeling site. Move labelers without rolls to installation site piece by piece diagonally. Consider all points for "Placing the Labeler" on page **34.**
3	Adjust the labeler with an air lever by the adjustable feet horizontally (if the labeler has rollers, please arrest the handbrake).
4	Consider when positioning the labeler that the applicator (extended tamp) may come close to the product up to 10-50 mm, but is not allowed to touch it.

Connecting the Labeler

Safety Instructions

⚠ CAUTION

Danger of drawing in by rotating elements.

DANGER OF DRAWING IN

Rotating elements at the labeler, rewinder, web brake and rollers of the friction unit.

- When working on the labeling system, disconnect from power supply.
- Keep away from moving parts.

NOTICE

Danger of property damage

The Alpha Compact can be connected to 100V up to 240V / 47-63 Hz. A false adjustment of the input-voltage leads to a damage of the system and is <u>no case of warranty!</u>

NOTICE

Danger of Failures

A false configured input leads to failures of the sensor but not to a damage of the controller.

NOTICE

Engine (motor) and controller have to be grounded. (grounding screw next to motor plug). Short time overloads influence the internal direct-current voltage in a negative way and should be avoided.

Connection to Supply Voltage

The Labeler needs electricity for its functions. Please find more details in the chapter “Technical Data”.

Requirements

- Power supply according to "Technical Data" is installed close (max. 1,5 m away) to the labeling site.
- Power voltage cable and net connection cable (s. in the accessory carton”)

Instruction

Please connect the labeler with supply voltage as follows.

Step	Procedure
1	Plug the power voltage cable of the system.
2	Connect the power voltage cable with supply voltage (standard 230 Volt socket).

Configuration Interfaces

Dependent on machine configuration the ALPHA Compact has different connections. The connections are located laterally at the cabinet of ALPHA Compact.

Belated extension of the interfaces is only allowed to be carried out by qualified personnel and is described in the service manual.

If optional connections are part of the scope of the delivery, the configuration of the inputs will be adjusted accordingly.

(X2) Product Sensor 1

The connection of the product sensor happens via a M12, 5 pin plug. The product sensor serves for the detection of the passing products and initiates the applying cycle.

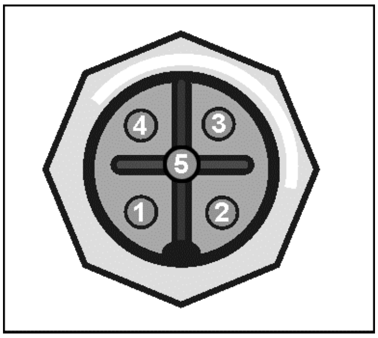


Fig.: 6-2 Pin Assignment Product Sensor

PIN	Assignment
1	+ 24V DC (BROWN)
2	NOT USED
3	GND (BLUE)
4	INPUT SIGNAL (BLACK)
5	NOT USED

Configuration Product Sensor Wenglor LD86PTC (Option)

The functioning of the standard sensor Wenglor LD86PCT3 is adjusted directly at the sensor via teach-button.

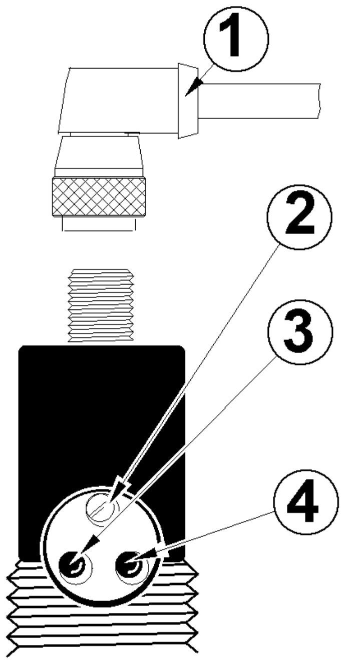


Fig.: 6-3 Wenglor LD86PTCT3

No.	Description
1	SENSOR-CONNECTION LINE
2	TEACH-BUTTON
3	LED RED DIRT STATUS DISPLAY
4	LED YELLOW SWITCHING STATUS DISPLAY

Flashing Teach-Button.	NCC/NOC	Teach Mode
1x	NOC (normally open contact)	Normal Teach
2x	NOC (normally open contact)	Minimal Teach
3x	NCC (normally closed contact)	Normal Teach
4x	NCC (normally closed contact)	Minimal Teach

If the sensor LD86PCT3 is used as product sensor with leading edge detection, you will have to adjust the teach mode **NCC (normally closed contact) normal teach.** If the sensor is used as low label warning, you will have to adjust the teach mode **NOC (normally open contact) normal teach**.

Requirements

- Labeler is connected with supply voltage.
- Control of product supply and if necessary the customer`s conveying system.

Instruction

Please adjust the sensor LD86PCT3 to the adequate teach modes.

Step	Procedure
1	Push the teach-button for 10 seconds until the LED changes from fast flashing to slow flashing.
2	Each pushing of the teach-button switches to the next teach-mode. The teach-modes are indicated by differing flashing of the shift indicator.
3	If the teach-button is not pushed for 15 seconds, the shift indicator goes back to normal indication mode and the settings are saved.

You will find further information on sensor settings in the manufacturer`s manual.

Product Sensor Positioning

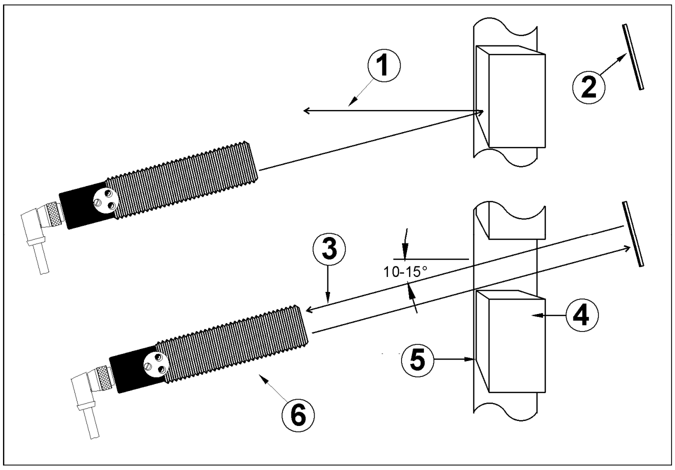


Fig.: 6-4 Light Barrier

No.	Description
1	NOT INTENDED REFLECTION OF PRODUCT SURFACE
2	REFLECTOR
3	REFLECTED LIGHT BEAM
4	PRODUCT
5	CONVEYOR BELT
6	LIGHT BARRIER

The fixing of the product sensor has to be performed before adjusting the sensitivity. Please mount the product sensor the way that you still can change adjustment later.

Requirements

- Product supply is stopped.
- Labeler is switched-on.
- Labels are loaded.

Instruction

Please position the product sensor as follows.

Step	Procedure
1	Please place a product below tamp in the way that a label would be applied on the desired position.
2	Now position the light barrier at the leading or trailing edge of the product according to the chosen configuration and connect sensor with controller via sensor connecting line.
3	Switch-on conveyor belt.
4	The labeler should be ready waiting for applying trigger. Let a product pass the labeler which should now apply a label. If the desired label position is not yet reached, please adjust it by moving the product sensor (and if necessary, the reflector). The position adjustment „trigger delay“ via controller should only be used, if the necessary product-sensor-position could not be reached because of space limit.

Sensitivity Adjustment

The following description deals with a light barrier with reflector [type: Wenglor **LD86NCT3**] and a reflective light sensor [type: Wenglor **HD12PTC**] (s. **Fig.:** 6-5). Please fix reflective light sensor or rather light barrier and reflector before you perform the following adjustments. You will find notices regarding correct positioning of the sensor on the following pages of this chapter. Perhaps the positioning of the sensor has to be adapted to the original products.

Subsequently the light barrier`s sensitivity adjustment as well as the light sensors` are explained because both variants are adjusted identically.

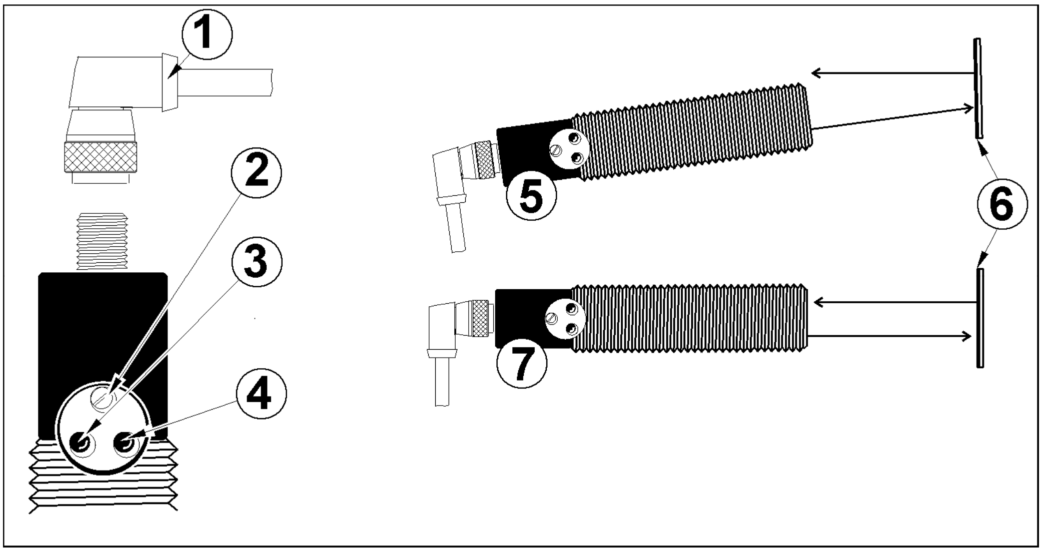


Fig.: 6-5 Adjustment Reflective Light Sensor

No.	Description
1	SENSOR-CONNECTION LINE
2	TEACH-BUTTON
3	LED RED DIRT STATUS DISPLAY
4	LED YELLOW SWITCHING STATUS DISPLAY
5	ALIGNMENT OF SENSOR / **NOT CORREKT**
6	REFLECTOR AT LIGHT BARRIER; RESP: PRODUCT AT LIGHT SENSOR
7	ALIGNMENT OF SENSOR / **CORRECT**

Requirements

- Product supply is stopped.
- Labeler is switched-on.

Instruction

Please adjust the sensitivity of the light barrier or rather the reflective light sensor as follows.

Step	Procedure
1	Please place a product below tamp in the way that a label would be applied on the desired position.
2	Now position the light barrier at the leading or trailing edge of the product according to the chosen configuration and connect sensor with controller via sensor connecting line.
3	Now adjust the light barrier to the reflector or rather the reflective light sensor to the product. Rotate the adjusting screw carefully clockwise up to mechanical stop. If the sensor detects the reflected (infrared) light, the switching status display LED (ON-LED) will switch on.
4	Pay attention that the red LED (fouling message) **does not** switch on. The following points could be the cause for the activation of the ERR-LED: • Incorrect distance to product/reflector • Fouling of sensor • Incorrect installation • Short circuit • Aging of sensor diode • Insecure area of operation
5	If the adjustment does not success, you will please repeat the steps up from procedure 4.
6	Examine again the sensor fixing. It has to be absolutely tightened and vibrations-resistant.
Only valid for the adjustment of a light barrier:	
7	Now place a product between light barrier and reflector. The LED should now go off. If the LED does not switch off, a light reflection will be produced by the product surface.
8	When you place a product again in front of the light barrier, the LED has to switch off.

If you have problems with false pulses because of reflections, you will have to adjust an adequate trigger blank. Or you should not adjust the light barrier and the reflector right-angled to the conveying system, but position them diagonal approx. 10-15°. Because of the diagonal positioning you avoid that light reflections from the product surface can reach the light barrier. In case of large distances between light barrier and reflector additional or bigger reflectors will enhance the switching reliability.

Test Run

After you are finished with all adjustments, please check the repeated application accuracy.

Requirements

- Control of product supply.
- Labeler is switched-on.
- Position of product sensor is adjusted.
- Sensitivity of product sensor is adjusted.

Instruction

Please check the repeated application accuracy as follows:

Step	Procedure
1	Pass, if possible, always the same product to the labeler to examine the application accuracy of the on each other pasted labels.
2	Examine the application accuracy of the on each other pasted labels.
	If the label position is not constant, you will please have to check the adjustments of the labeler (from the beginning of this chapter) and the product sensor.

(X3) Low Label (Option)

The connection of the Low Label Sensor happens via a M12, 4 pin plug socket. This plug socket is suitable for the optional Low Label Warning 35001817.

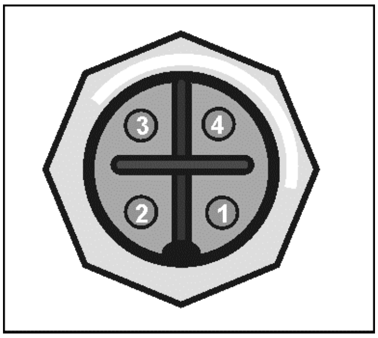


Fig.: 6-6: Pin Assignment Low Label Warning

PIN	Assignment
1	+ 24V DC (BROWN)
2	NOT USED
3	GND (BLUE)
4	INPUT SIGNAL (BLACK)

Low Label Sensor (Option)

The low label prewarning is used to display a pending end of the label roll. If a pending roll end was displayed, the user should insert a new label roll to keep the labeler ready for operation.

If a particular roll diameter is underrun, a warning message is issued by a signal at the output X6 alarm lamp (yellow). The labeler is ready for operation as long as there are no labels anymore available. Then the labeler changes to failure and stops.

For the optional low label prewarning (article number 35001817) a mounting has to be fixed at the unrolling arm of the sensor. The sensor has to be connected at the connector **X3 Low Label**.

An alarm lamp can be connected optionally at the socket **X6 Alarm lamp**.

If the low label prewarning is not detected after installation, a Service Technician should check in the configuration menu if the parameter 004 at +16 (Function: Store of low label signal) and parameter 045 at +64 (Low-paper Sensor) are selected.

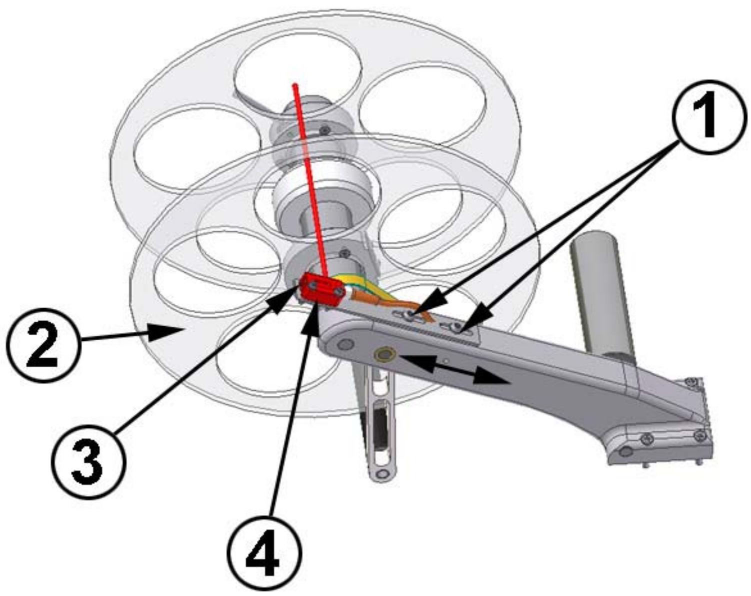


Fig. 6-1: Low label prewarning

No.	Assignment
1	SCREWS FOR ADJUSTMENT
2	TABLE DISC
3	SENSOR
4	RANGE OF HEADLIGHTS‘ ADJUSTMENT

Requirements

- Product transport is stopped.
- Labeler is ready for operation.

Required equipment:

- 4mm Hexagon socket wrench
- Philips screw driver Große 00

Instructions

Please adjust the Low Label Sensor as follows.

Step	Procedure
1	Turn the table disc so that the light beam of the sensor lights between the gaps.
2	Turn with the Philips screw driver the range of headlights adjustment counter clock-wise down until the label roll is not detected anymore by the sensor (display LED goes out).
3	Turn the headlights adjustment clockwise appr. 1 rotation until the label roll is safely detected by the sensor. Observe that no objects are detected in the background by the sensor.
4	Loosen the adjustment screws with the hexagon socket wrench. Turn therefore the screw counter clockwise.
5	Adjust the sensor so that the light beam of the sensor detects the label roll's end. The farther you move the sensor to the roll's beginning, the earlier the low label prewarn-ing is carried out.
6	Tighten again the screw for adjustment clockwise.

(X5) Encoder (Option)

The connection of an encoder is arranged by a M12, 5 pin plug and enables an automatic adaption of the label applying speed at different or variable product transportation speeds.

For the initial operation of the encoders a HMI Display or an *"USB-connection is required. When using the encoder, you have to consider that no entries at HMI panel or HMI display are possible during a label applying procedure. This is also valid if the label application was interrupted by a stop of the conveying system. Furthermore, a new application cycle is only started if the start trigger signal was reset before the new applying cycle. Thus the products have to be fed separately.

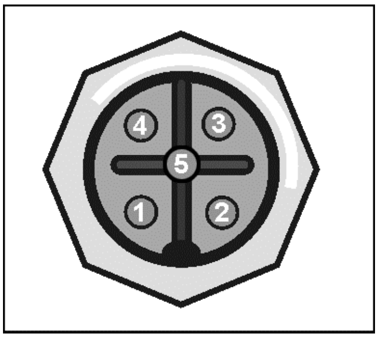


Fig.: 6-7: Pin-Assignment Encoder

PIN	Assignment
1	+ 24V DC (BROWN)
2	NOT USED
3	GND (BLUE)
4	INPUT SIGNAL (NPN OR PNP DEPENDING ON JUMPER POSITION)
5	NOT USED

Align Encoder (Option)

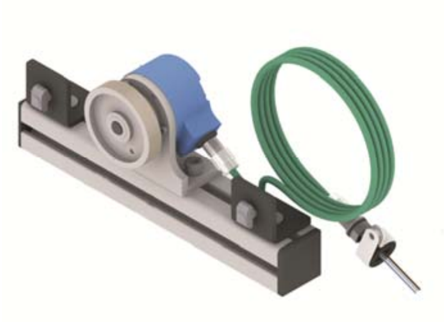


Fig. 6-2: Encoder with friction wheel

For an optimum labeling result the label applying speed has to coincide with the product transportation speed.

If the products are always fed with the same speed to the labeler Alpha Compact, a fixed adjustment of the applying speed is sufficient. If the product transportation speed varies, the label applying speed has to be synchronized by means of an optional encoder. Generally the Alpha Compact can be used with most encoders. For the initial operation of the encoder, a HMI display or an *"USB-connection is required.

To be able to dimension the encoder and the transmission correctly, please find below the procedure for our standard encoder with 2000 pulses per rotation (article number 33003611). For the initial operation of the encoder, an HMI display or an *“USB-connection is required.

Requirements

- HMI diplay or *^{2}USB-connection is available
- Labeler is turned on.
- Encoder is installed and connected at the labeler.
- Possibility for start-triggering.

Step	Procedure
1	Investigate the pulses per rotation of the encoder. (Our standard encoder article number 33003611 provides **2000 pulses per rotation**.)
2	Investigate the diameter of the used friction wheel. (Our standard frictions wheel provides **63,6mm** diameter (article number 61801017). (*If the encoder is directly installed at a spindle of the customer's conveying system, please use the spindle's diameter.)
3	Calculate the values for programming 158 and 159 in programming mode:

Design of parameter 158 and 159:

$$\text{Basic parameter} = 799 \times \frac{\text{Pulses per rotation } [\mathbf{2000\ /\ Umd}.]}{\text{Friction wheel diameter } [\mathbf{63{,}6\ mm}]} = \mathbf{25125}$$

The basic parameter may not be larger than 65535. Appropriate combinations from encoder and friction wheel may not be connected.

$$\text{Programming 158} = \frac{\text{Basic parameter } [\mathbf{25125}]}{256} = \mathbf{098}$$

Programming 159 = Basic parameter [**25125**] – (Value programming 158 [**098**] x 256) = **037**

Step	Procedure
4	Enter in programming mode* in programming 158 the calculated value. (for our standard the value **098**).
5	Enter in programming mode in programming 159 the calculated value. (for our standard the value **037**).
6	Enter in programming mode programming 056 the value +64. (for our standard with doubled brake ramp +004 + 064 = **068**).
7	Enter in programming mode programming 052 a value for the manual label application a speed (e.g. 20 M/min).

*The password for programming mode is 0123.

(X6) Alarm Lamp (Option)

The connection of an optional alarm lamp happens via M12, 8 pin plug.

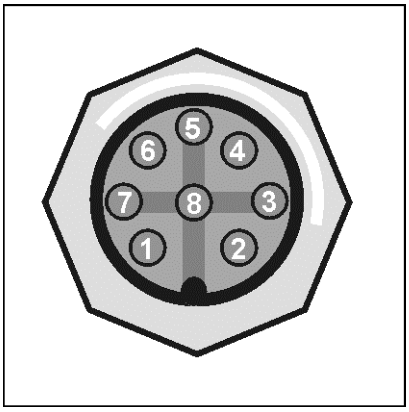


Fig.: 6-8: Pin Assignment Alarm Lamp

PIN	Assignment
1	+ 24V DC RED LAMP
2	+ 24V DC YELLOW LAMP
3	+ 24V DC GREEN LAMP
4	NOT USED
5	GROUND (-)

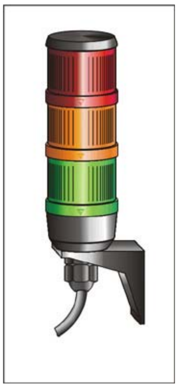


Fig.: 6-9 Alarm Lamp

Lamp	Description
Red (permanent light)	Flashes in case of system failure / error (READY-Signal is re-set). Possible cause: label out or labeler error.
Yellow (permanent light)	Low Label Warning
Green (permanent light)	System ready (Ready Signal is set)

(X7) Label Sensor

The connection of the label gap sensor happens via a short cable that is lead out laterally of the cabinet and that has a M8, 5 pin plug socket. This socket is suitable for the standard provided label gap sensor.

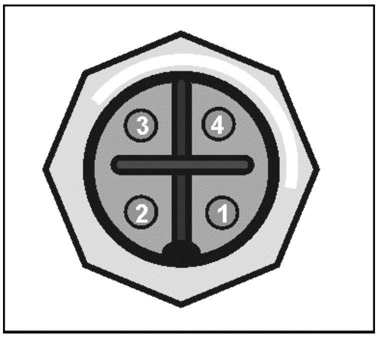


Fig.: 6-10: Pin Assignment Label Sensor

Nr.	Bezeichnung
1	+ 24V DC (BROWN)
2	NOT USED
3	GND (BLUE)
4	INPUT SIGNAL (BLACK)

Positioning of Label Sensor

If the detection of the label front edge is problematic, the photocell will have to be moved to a preferably free field (if printed) in the label. So an incorrect measurement of the photocell can be avoided. In case of a form-stamped label the photocells have to be moved to the “straightest” label edge.

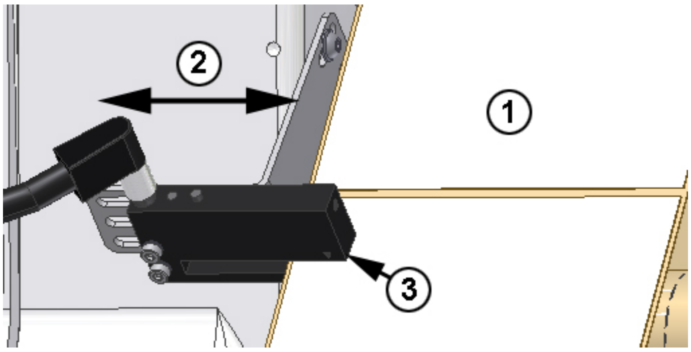


Fig.: 6-11 Label Sensor Adjustment

No.	Description
1	LABEL WEB
2	DIRECTION OF THE ADJUSTMENT
3	LABEL SENSOR

Required Tools

- 2,5 mm setscrew wrench

Requirements

- Labels are loaded.
- Product supply is stopped.
- Labeler is separated from supply voltage and air pressure.

Adjusting Label Sensor to Label

Instruction

Please adjust the label sensor position as follows.

Step	Procedure
1	Loosen both screws below the sensor bracket.
2	Move the sensor to desired position and tighten both screws again.
3	Arrange then a label calibration.

A change in position of the sensor in longitudinal direction is generally not permitted and would make necessary an adjustment in service program.

Scale label sensor to label

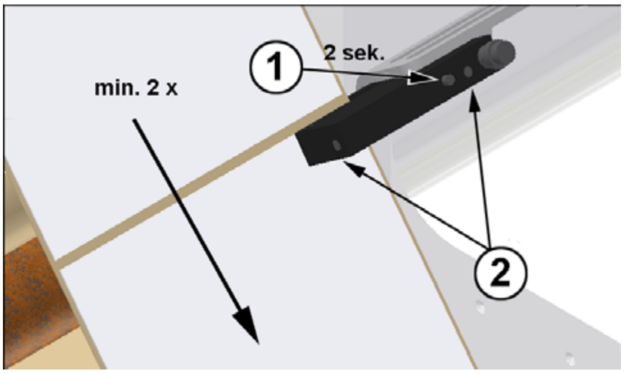


Fig.: 6-12 Scale labels

No.	Description
1	TEACH-IN BUTTON
2	LED DISPLAY

If you push the teach in button longer than 6 seconds, you will switch the light-/dark detection (NO/NC) at the sensor. To scale the label, push the teach in button only for 2 seconds.

If the labels are longer than 300mm, the label length has to be adjusted in configuration menu in parameter [01Label Length].

Requirements

- Label web is inserted.
- Product feed is stopped.

- Labeler is switch on.

Instruction

Please adjust the label sensor to the product as follows.

Step	Procedure
NOTICE	**Push the teach button not too long. You will misalign the basic settings of positive / negative edge.**
1	Push the teach-button for 2 seconds until the LED display flashes permanently.
2	Pass the label web through the forked light barrier within the next 2-8 seconds. At least 2 labels have to be passed through the forked light barrier. The LED display signalizes the teach-action by fast flashing.
3	After teach-action the LED display informs by flashing about the result **2x flashing**, teach-action is finished successfully. **4xˆflashing**, teach-action is not finished successfully. In this case please repeat teach-action.
4	Please make a label calibration (see p**. 64**).

Option micro-switch scanning as product sensor

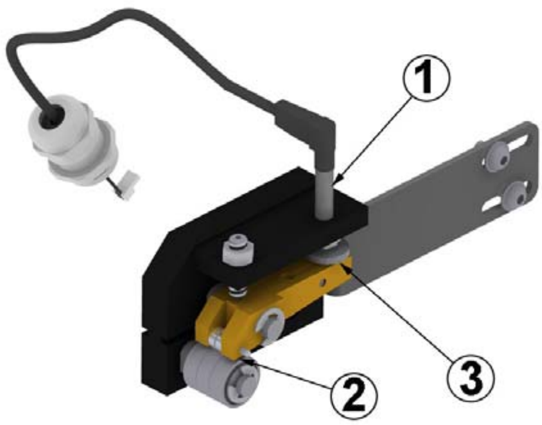


Fig.: 6-13: Micro-switch

No.	Description
1	SENSOR
2	LABEL DETECTION POSITION
3	KNURLED WHEEL

The optional micro-switch scanning is used at high gloss or transparent labels that are not detected by the standard label sensor.

Requirements

- Label web is inserted.
- Product feed is stopped.
- Labeler is switch on.

Instruction

Please adjust the micro-switch scanning as follows.

Step	Procedure
1	Lead a label gab at the label scanning position.
2	Turn the knurl wheel until the indicator light of the senosor goes straigt out.
3	Press the [Start] button for checking the adjustment. If necessary, correct the setting by small adjustment on the knurl wheel.

(X8) Applicator (Option)

The applicator connection happens via M12, 5 pin socket. This enables the labeler`s connection with further optional sub-assemblies.

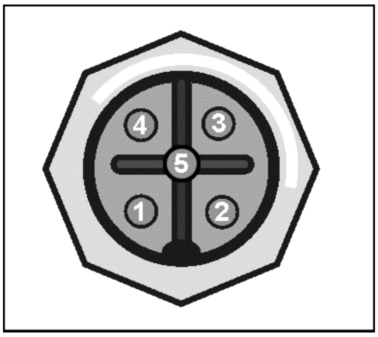


Fig.: 6-14: PIN Assignment Applicators

PIN	Assignment
1	+ 24V DC (BROWN)
2	NOT USED
3	GND (BLUE)
4	NOT USED
5	NETWORK

(X9) I/O (Option)

The I/O interface serves for the connection of the Alpha Compact with new machine options or the customer`s controller. The PIN 4 is not used for the external trigger. It has the same function as the [Start] button of the Control Panel. A signal will be activated and maybe failure messages will be reset by pushing.

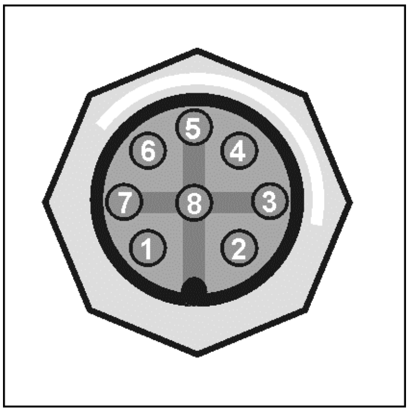
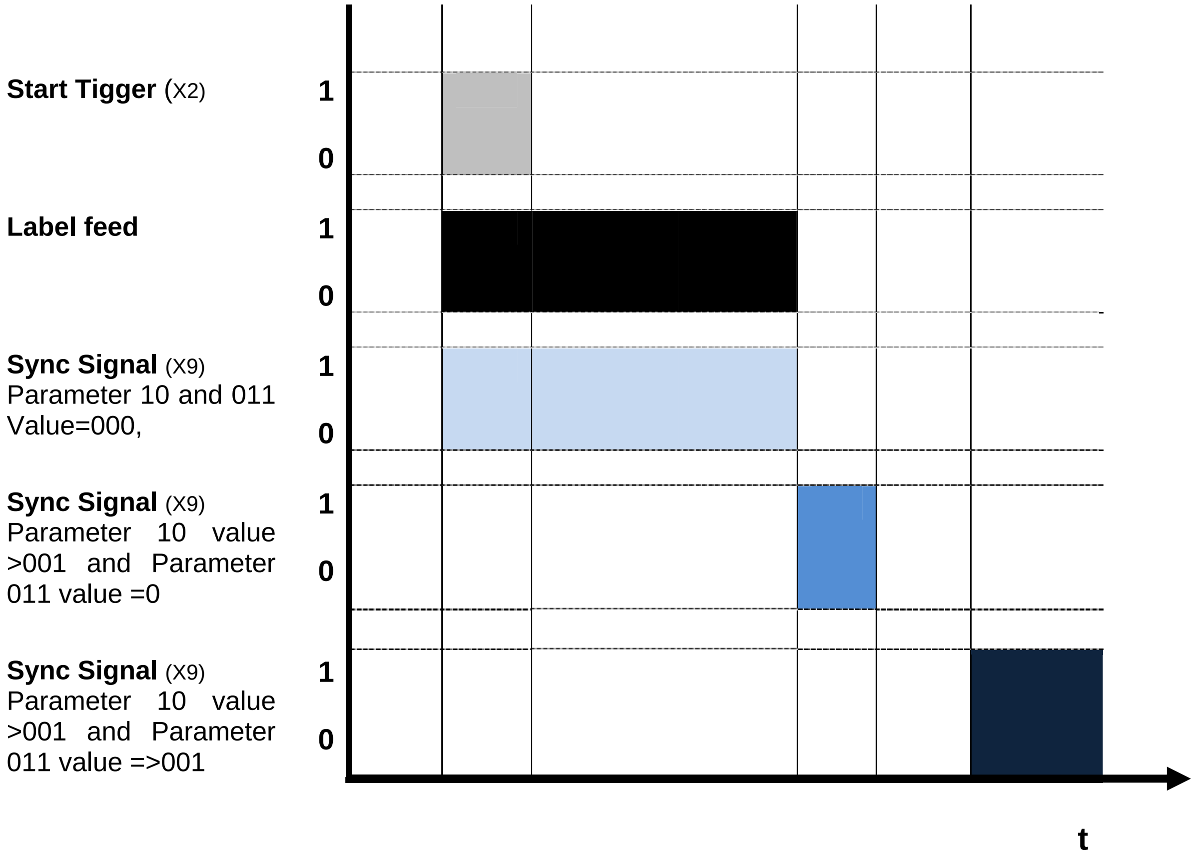


Fig.: 6-15: PIN Assignment X9 I/O

*All outputs which are supplied via (P 24) are not alowed to consume together more than max. 500mA.

PIN	Assignment
1	OUTPUT SIGNAL: **READY**, (N.O.) MAX: 200MA, MAX. 30V
2	OUTPUT SIGNAL: **LOW LABEL WARNING** (N.O.) MAX. 200MA, MAX. 30V
3	OUTPUT SIGNAL: **SYNC**, (N.O.) MAX: 200MA, MAX. 30V
4	INPUT SIGNAL: **REMOTESTART** MAX. 10MA, LOW < 3VDC, HIGH > 15VDC
5	COM INPUT SIGNAL
6	COM OUTPUT SIGNAL
7	+24V (P24)
8	0 V TO EARTH

Following graphics shows the correlation between the Start Trigger Signal and the SYNC-PULSE TIME depending on parameter setting 010 and 011 in configuration menu (see p. 97).



Connection example

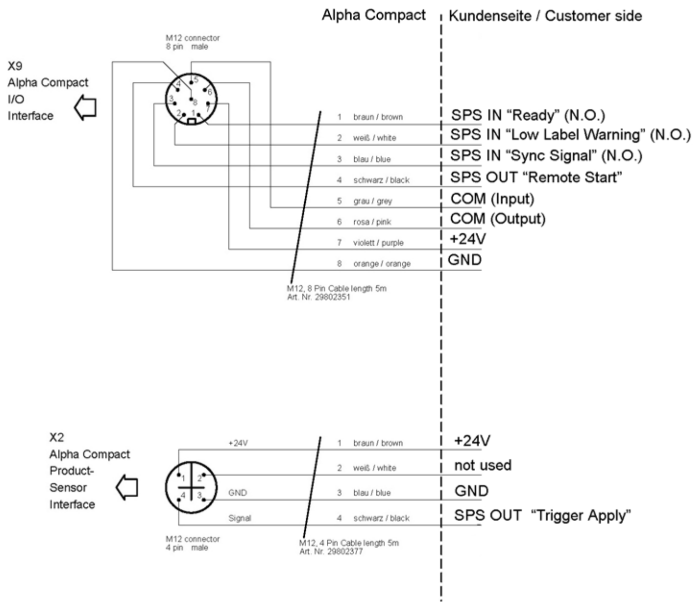


Fig.: 6-16: Connection example

(X10) HMI

NOTICE

Danger of damaging.

The labeler may be damaged when separating or connecting the HMI when the labeler is turned on.

- Separate or disconnect the HMI only in turned-off condition.

A simple user panel or a display panel with a M12, 5 pin plug may be connected at the Alpha Compact. Both controllers are connected with the Alpha Compact with the same connection. When using an optional USB-connection, the settings can also be configured by a PC. Depending on adjustment of the Alpha Compact, a controller is not required anymore after initial operation. To avoid parameter adjustments arranged by unauthorized personnel, the controllers can be disconnected by the labeler. For further information on operator's panel please see p**. 79**

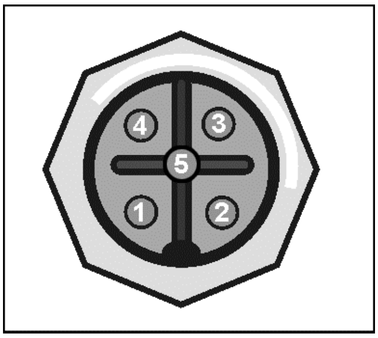


Fig.: 6-17: PIN assignment applicators

PIN	Assignment
1	+ 24V DC (BROWN)
2	NOT USED
3	GND (BLUE)
4	NOT USED
5	NETWORK

USB connection (Option)

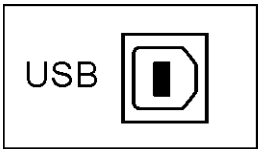


Fig.: 6-18: USB connection

The Alpha Compact can optionally be equipped with an USB 2.0. The Alpha Compact will be programmed and operated with software and PC. Additionally extensive diagnosis and error detections can be evaluated. A controller is not necessary in this case.

Adjust labeler to label liner width

Safety instructions

CAUTION

Danger of drawing in by rotating elements.

DANGER OF DRAWING IN

Rotating elements at labeler, rewinder, web brake and rollers of the friction unit.

– Disconnect labeler from power voltage when performing any work.

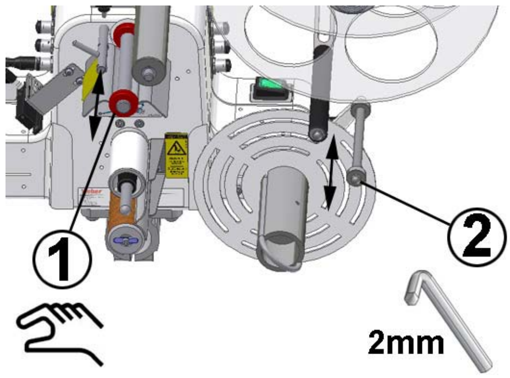


Fig.: 6-19: Adjust labeler to label liner width

No.	Description
1	GUIDE RING
2	ADJUSTING RING

The labeler has to be adapted to the guiding of the web brake and to the dancer arm of the unwinding in the label liner’s width. There are two guide rings at the guiding of the web brake that can be moved manually. At the dancer arm of the unwinding there are two adjusting rings that are secured by a setscrew. Generally is the position of the label to the cabinet fixed. To adjust the label width, only the guide ring and the adjusting ring to the open outside are adapted. The guidings have to be adjusted so that between label liner web and guiding there is a gap of appr. 0,1 mm.

Requirements

- Labeler is without power and compressed air
- Label rolls

Required equipment

- 2mm Hexagon socket wrench

Please adjust the labeler to the label liner's width as follows.

Step	Procedure
1	Insert the label liner web in running direction between the guide rings to the web brake.
2	Move the outer guide ring until the label liner web has a clearance of appr. 0,1 mm to the guide rings.
3	Loosen the setscrew at the outer adjusting ring by turning the setscrew counter clockwise. Please use therefore a 2 mm hexagon socket wrench.
4	Insert the label liner web in running direction between the adjusting rings of the dancer arm. Move the outer adjusting ring until the label liner web has a clear-ance of appr. 0,1 mm to the adjusting rings.
5	Tighten again the setscrew at the outer adjusting ring by turning it clockwise.
6	Remove the label liner web.

Loading and Changing Label Material

Safety Instructions

CAUTION

Danger of drawing in by rotating elements.

DANGER OF DRAWING IN

Rotating elements at labeler, rewinder, web brake and rollers of the friction unit.

- Disconnect labeler from power voltage when performing any work.

NOTICE

Danger of damage.

The rewinder has a free-wheel that avoids the winding up of the label liner material. It can be damaged by forcibly turning against the wind-up. Do never turn the rewinder against rewind direction.

Label web guiding for Wipe-On Mode

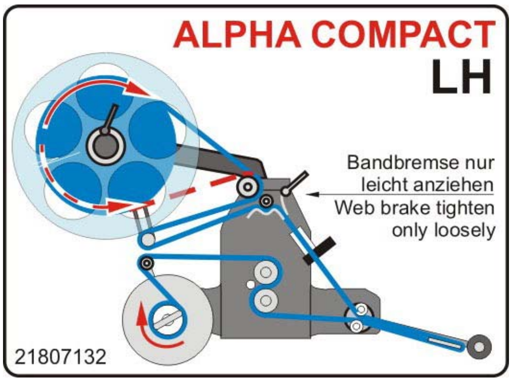


Fig.: 6-20 Loading Scheme LH

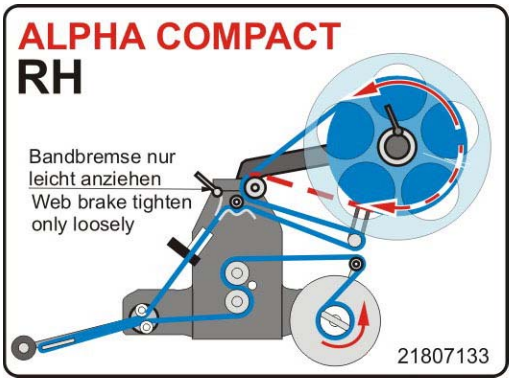


Fig.: 6-21 Loading Scheme RH

The threading scheme is sticked onto the labeler. For a better understanding, it shows the label web without traverse, mouning or hand guard. If the labeler has optional special equipment, an appropriate threading scheme is sticked onto the labeler.

Requirement

- The labeler has to be switched off when loading label web.

Instruction

Please load label web as follows.

Step	Procedure
1	If label web is still loaded, cut it in region of the peeler blade.
2	Open the clamp lever of the unwind disc and take it off. If there is still a label roll at the unwind mandrel remove it.
3	Load the label roll onto the unwind mandrel which should be central on the unwinder`s axis. (If not, loose the set screw to the unwind mandrel and align it. Afterwards fix the new position with the set screw.
4	Lay the label material around the deflection roller.
5	Lay the label material around the dancer arm roll and to front side. Release the web brake by pushing the dancer arm.
6	Set the unwind disc onto the unwind axis and push it at the label roll.
7	Fix unwind disc with clamp lever.
8	Guide the label material between web-leading-axis and web-leading-plate below the web brake.
9	Adjust the red web-leading-rings in a distance from < 1mm to outer paper edge
10	Release web brake with the help oft he clamp lever so the paper can be loaded easier. Adjust the web brake on to a low brake level.
11	Guide the label material below the upper leading roll to the peeler blade.
12	Guide the label material around the peeler blade.
13	Pass the label material above the lower leading roll back to the friction roll.
14	Remove all labels at the first 600 mm of the label web (from peeler blade) and put the label liner around the friction roll.
For an easy loading the friction roll can be lifted by a rounded spindle.	
15	Guide the label liner via the friction roll to the rewinder.
16	If necessary remove the label liner roll at the rewind core. Therefore remove the web clamp and take out the label liner roll. Fix the new loaded label material via web clamp U-shaped on the core.
17	Guide the label web to the rewind core and fix the new loaded label web via web clamp U-shaped on the core.

Dancer arm unwinder- setting

Safety instructions

⚠CAUTION

Danger of drawing in by rotating elements.

DANGER OF DRAWING IN

Rotating elements at labeler, rewinder, web brake and rollers of the friction unit.

– Disconnect the labeler from supply voltage when performing any work.

NOTICE

Danger of production disruptions.

The setting of the brake belt for the label unwinder may only be arranged by trained personnel!

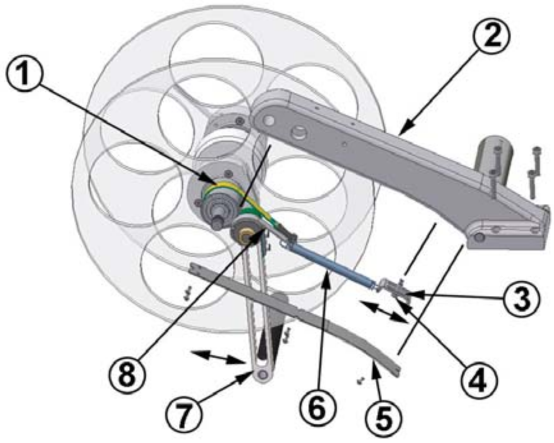


Fig.: 6-22: Unwinder in dismantled view

Schritt	Vorgehen
1	BRAKE BELT OUTSIDE
2	UNWINDER ARM
3	ADJUSTMENT TO TENSION SPRING INSIDE
4	SCREW TO CLAMPING OF TENSION SPRING INSIDE
5	COVER PLATE
6	TENSION SPRING INSIDE
7	DANCER ARM WITH DEFLECTION ROLLER
8	BRAKE BELT INSIDE

The unwinding is used to support the label roll. The label material is peeled off by a drawing unit from the unwinder. To provide a constant label web tension, the unwinder axis slows down or loosens if applicable by two brake belts that are triggered by a dancer arm.

The brake belts are preset ex-factory. **This setting is suitable for most applications**. Only particular applications (e. g. with slightly/heavily labels, High-Speed applications aso.) or by long-term operation (material fatigue of spring tension) a correction of this setting may become necessary.

If the web tension is too low, the unwinder brake is not able to brake the label roll. The label roll runs after each label feed so that an exact applying is not possible. If the dancer arm tension is too high, the stepper motor has to work permanently against the brake power. This will lead to a motor overload or cause errors of positions of the applied labels on the product.

The adjustment of the web tension is arranged by the adjustment to the spring tension inside. It is located at the unwinder arm. To get access to it, please dismantle a cover plate at the bottom side of the unwinder arm.

Requirements

- Label web is inserted.
- Product feed is stopped.
- Labeler is ready for operation but stopped.

Instructions

Please adjust the tension spring for dancer arm tension as follows.

Step	Procedure
⚠CAUTION	**Caution! The tension spring may be tensioned!**
1	Dismantle the cover plate below the unwinder arm. Remove therefore the 5 screws.
2	Loosen the 2 screws slightly for clamping of the tension spring inside.
3	Move the adjustment to the tension spring inside slightly forward or backward depending on demand. (The clamping screws are ex-factory centered in long hole).
4	Tighten again the 2 screws for clamping.
5	Arrange several sample labelings to check the adjustment.
6	If the adjustment was successful, please mount again the cover plate. If the adjustment was not sufficient, please return to step 2 and correct it.

The dancer arm should in resting position also be slightly tensioned. Generally it is valid: the faster the feeding speed, the higher the spring power has to be.

Label Calibration

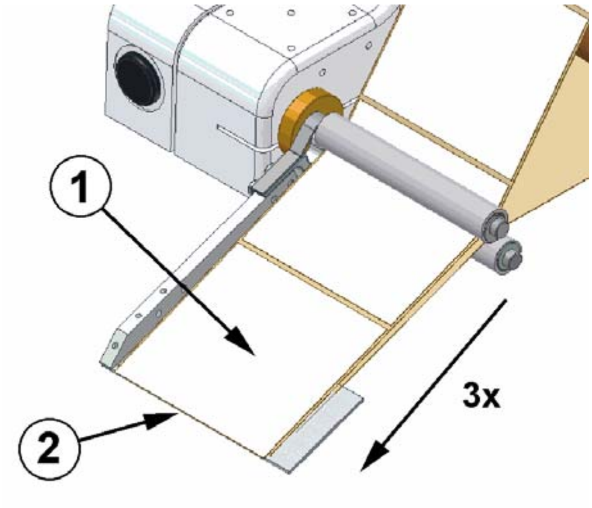


Fig.: 6-23: Calibrate labels

No.	Description
1	LABEL
2	PEELER BAR

The Alpha Compact provides a function to calibrate labels. This function enables the labeler to detect the label length and the position to the peeler bar. After the calibration, the label is located flush with the peeler bar. For most applications, this is the correct label position.

If a different label position is required, it can be adjusted with the potentiometer [Adjust] when connecting the HMI Panel (in standard configuration) s page 79. If a HMI Display is connected to the labeler, in functions menu [002 LABEL POS.] the label position can be adjusted.

Requirements

- Labels are loaded.
- Product supply is stopped.
- Labeler is connected with supply voltage.

Instruction

Please calibrate the labels via control panel as follows.

Step	Vorgehen
CAUTION	**Danger of drawing in by rotating elements.**
1	Push the [Cal] button. The labeler performs 3 applying cycles for the calibration. Afterwards the leading edge of the label is flush to the peeler bar.

Instruction

Please calibrate the labels via display controller as follows.

Schritt	Vorgehen
CAUTION	**Danger of drawing in by rotating elements.**
1	If you are not in the starting mode please push the [Start] button.
2	Push the buttons [◀] + [▶] together at the same time. The labeler performs 3 applying cycles for the calibration. Afterwards the leading edge of the label is flush to the peeler bar.

Settings at Wipe-On System

Safety Instructions

CAUTION

Danger of drawing in by rotating elements

DANGER OF DRAWING IN

Rotating elements at labeler, rewinder, web brake and rollers of the friction unit.

- Disconnect the labeler from supply voltage when performing any work.

CAUTION

Danger caused by moving products

DANGER OF BEING CRUSHED

*2 In case of a flexible peeler blade the passing product can push the peeler blade away.
*2 In case of an inflexible peeler blade objects can be crushed between product and peeler blade.

- Stop the supply of products.

Possibly a few adjustments are necessary until you will achieve the desired labeler result. In any case the products should not be decelerated too much during "labeling" and are never allowed to be touched by the peeler bar. All in all two settings are possible for the pusher roller of the labeler.

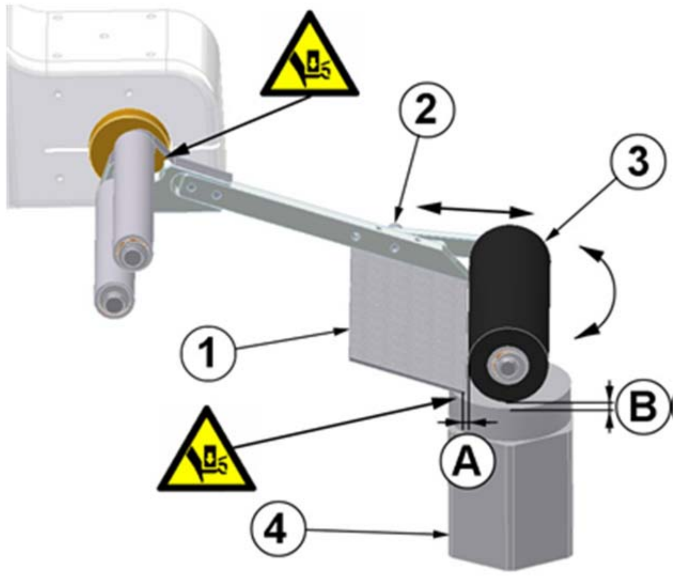


Fig.: 6-24 Wipe-On-Setting

No.	Description
1	PEELER BLADE
2	CLAMPING SCREW
3	PUSHER ROLLER
4	PRODUCT
A	DISTANCE TO PEELER BAR
B	DISTANCE BETWEEN PUSHER ROLLER AND PRODUCT

A the distance between peeler bar and pusher roller.

B the adjustment of the pusher roller to the product.

Requirements

- Labels are loaded.
- Supply of products is stopped.
- Labeler is adjusted in direction to the product.
- Labeler is separated from supply voltage.

Instruction

Please adjust the Wipe-On system as follows.

Step	Procedure
1	Please put a product directly below the peeler bar. The conveying system has to be switched-off.
2	Loosen the screw of the roller holder and adjust the roller approx.to product height. **Remark that longer labels always need a bigger distance of the pusher roller.**
3	Fix the screw and let a few products pass the labeler (without application) to check the adjustment.
4	Switch-on the labeler at on- /off-switch.
5	Please adjust the speed of label feed to the speed of products.
6	Make some test-applications and correct the application result if necessary (see p.123).

Adjust peeler blade to the deflection rollers

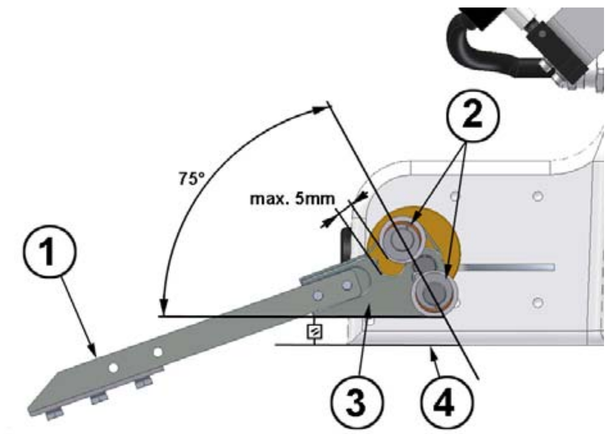


Fig.: 6-25: Adjust peeler blade

Nr.	Bezeichnung
1	PEELER BLADE
2	DEFELECTION ROLLERS
3	PEELER BLADE BRACKET
4	LOWER EDGE COVER

To decrease the danger of being crushed between peeler blade’s mounting and deflection roller, the gap may not be larger than 5 mm. To ensure this, an angle of 75 ° between peeler blade’s mounting and deflection roller has to be kept. In our factory, a mounting device is used for alignment. By means of this device, the peeler blade’s mounting is aligned parallel to the lower edge of the cover. If this factory setting is changed for adaption to the customer’s conveying system, the maximum distance of 5mm has to be reconstructed. To adjust the distance, at the bottom side of the cover, two clamping screws are loosened and then the extender bearing is turned into the appropriate position.

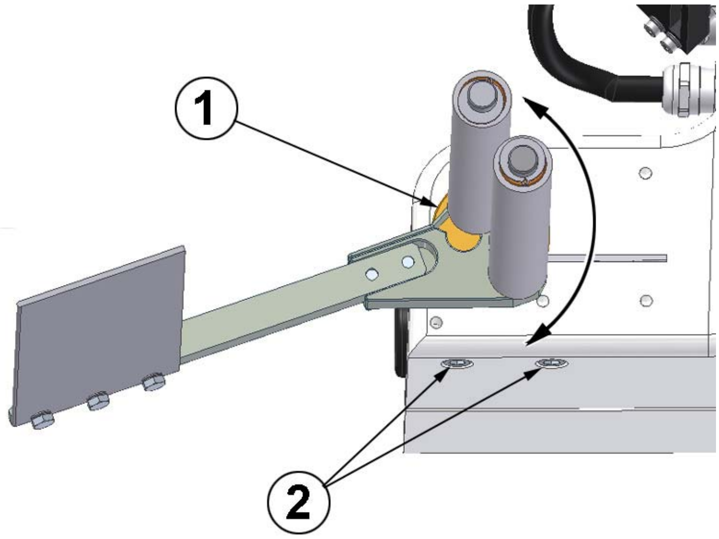


Fig.: 6-26: Adjust extender bearing

No.	Description
1	EXTENDER BEARING
2	CLAMPING SCREWS

Requirements

- Product feed is stopped.
- Disconnect labeler from voltage supply.

Required equipment:

- 5mm Hexagon socket wrench

Instruction

Please adjust the distance of the peeler blade's mounting to the deflection roller as follows.

Step	Procedure
1	Loosen the both screws (Fig.: 6-27, Pos.2) to clamping.
2	Turn the extender bearing to that the gap between peeler blade's mounting and deflection roller is not more than 5 mm.(Check the gaps distance (e.g. with a 4mm and 5mm hexagon socket wrench).
3	Tighten again the both screws (Fig.: 6-27, Pos.2) to clamping.

Settings coupling rewinder

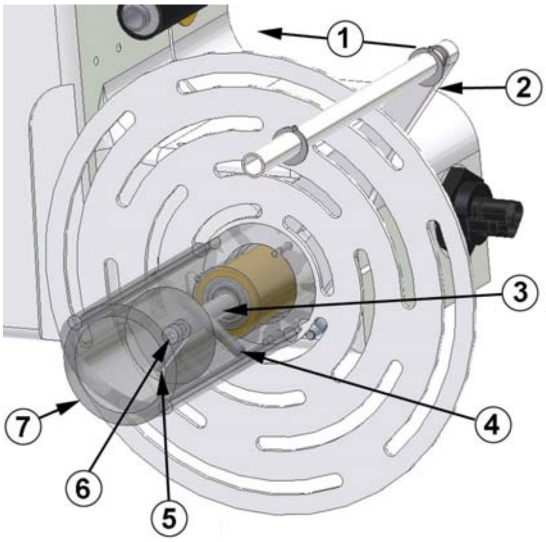


Fig.: 6-27: Adjustment coupling

No.	Description
1	DIRECTION OF ADJUSTMENT
2	DANCER ARM
3	CORE
4	ADJUSTMENT SCREW
5	CYLINDER BOLT
6	SETSCREW (FIXING)
7	AXIS TO COUPLING

NOTICE

Danger of damage.

The rewinder has a free-wheeling that impedes the unwinding of the liner material. By violent turning against the rewinding direction, it can be damaged. Never turn the rewinder against the rewinding direction. Do no push the dancer arm by force over its end position.

The coupling to the rewinder is aligned when the system is delivered. To avoid damages of the dancer arm due to overload, the dancer arm adjusts itself at inappropriate force.

Requirements

- Label web is not inserted.
- Product transportation is stopped.
- Labeler is turned off.

Required equipment

- 2,5mm and 3mm hexagon socket wrench

Instructions

Please adjust the coupling as follows.

Step	Procedure
1	Loosen the setscrew (Fig.: 6-27, Pos.6).
2	Loosen the adjusting screw (Fig.: 6-27, Pos.4).
3	Push the dancer arm (Fig.: 6-27, Pos.2) in direction of the adjustment (to the cover) and keep it pushed.
4	Tighten slightly the adjusting screw (Fig.: 6-27, Pos.4).
5	Release again the dancer arm (Fig.: 6-27, Pos.2).
6	Loosen the adjusting screw slightly (Fig.: 6-27, Pos.4) until the dancer arm moves slowly backward by the spring force.
7	If the dancer arm achieves an angle of appr. 45 degree to the cover, please tighten again the adjusting screw (Fig.: 6-27, Pos.4).
8	Tighten then finally the setscrew again (Fig.: 6-27, Pos.6).

*²Air assist setup-with optional applicator

CAUTION	**Danger due to actively controlled movements.**  DANGER OF BEING CRUSHED! The movements of the applicator are driven by pneumatic cylinders. - Maintain a distance from applicator.

For the label application the label must be pressed against the Tamp Pad until it is captured by the vacuum. This is arranged by the air jet of the air assist unit.

The air assist unit consists of an air assist tube (5) with three holes and one flow control valve (Pos.1), that may set the effusing air.

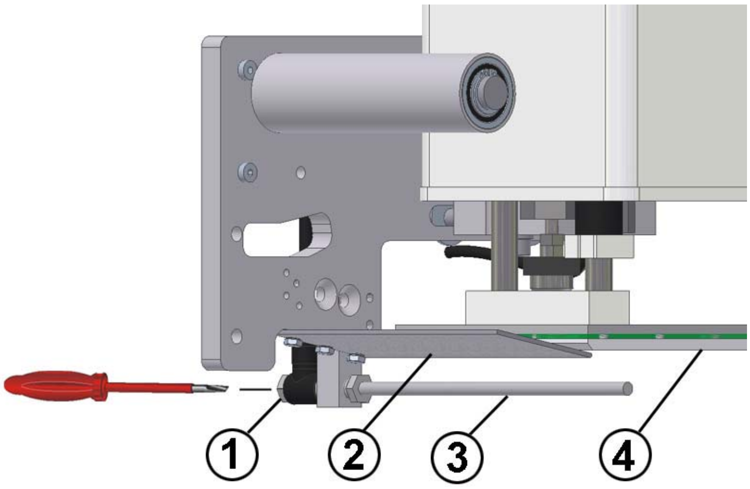


Fig.: 6-28: Air assist unit at applicator

No.	Description
1	FLOW CONTROL VALVE (AIR ASSIST)
2	PEELER BLADE
3	AIR ASSIST TUBE
4	TAMP PAD

*² only if system has the appropriate feature.

Requirements

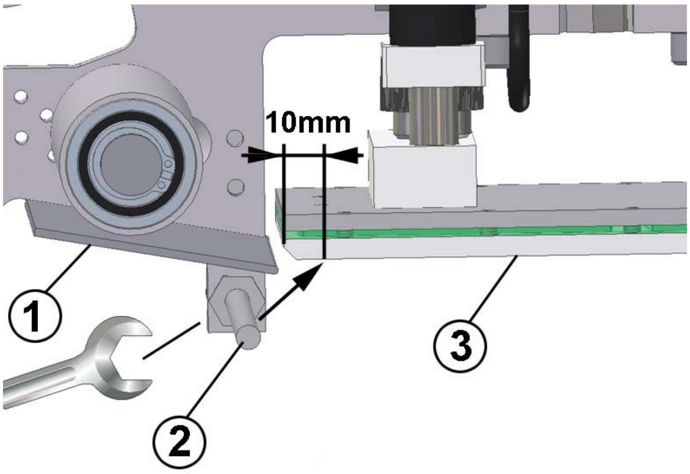
- Labeler is turned on
- No transportation of products

Required equipment:

- Screw driver G00
- Screw wrenchs 14 and 12

Instructions

Please adjust the air assist as follows

Step	Procedure
1	Check firstly if the air assist beams hits appr. 10 mm behind the leading edge of the tamp (see Fig.: 6-29).
	 Fig.: 6-29: Air assist unit (side view) **No.** / **Description** **1** PEELER BLADE **2** AIR ASSIST TUBE **3** TAMP PAD
2	If an adjustment is necessary, the counter nut has to be unscrewed and the tube has to be turned into position and then arrest the nut again.
3 CAUTION	Following adjustment can only be arranged during a label feed and required a lot of experience. DANGER OF BEING CRUSHED IN! Keep away from tamp.
4	Push the [ENTER]-button of the printer in order to feed a label. Regulate in the meantime the air beam by the flow control valve at the bottom side of the stainless steel cabinet (see p. 71). - turning clockwise leads to a weakening. - turning counter clockwise leads to an amplification of the air jet.
5	Repeat step 4 until the label is pushed safely against the tamp pad. Search for an adjustment requiring only a little quantity of air pressure. You will thus reduce the air pressure consumption and you will reduce possible air rotations.

*²Adjust angle of rotation at optional rotating tamp

CAUTION

Danger due to actively controlled movements..

DANGER OF BEING CRUSHED!

The movements of the applicator are driven by pneumatic cylinders.

- Maintain a distance from applicator.

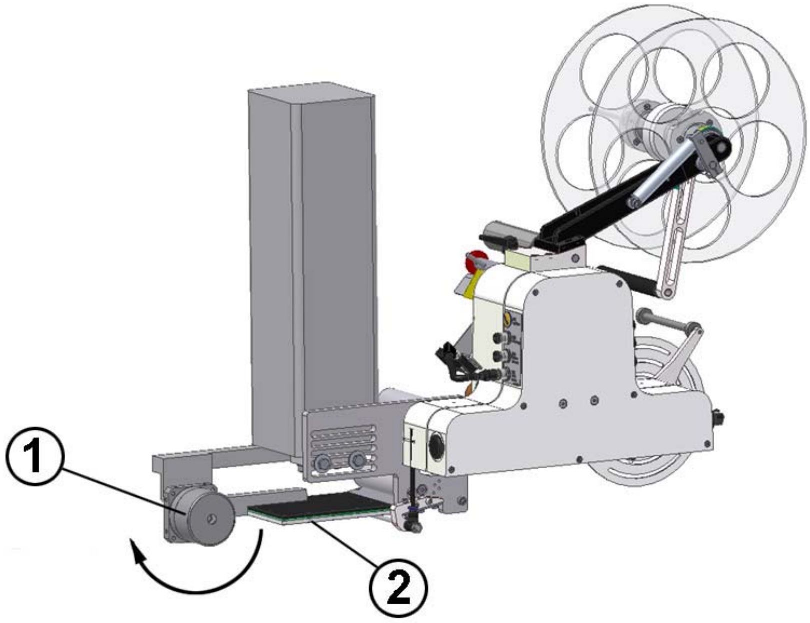


Fig. 6-3: Alpha Compact with rotating tamp

No.	Description
1	SWING MODULE
2	ROTATING TAMP

The rotating angle of the rotating tamp has to be adjusted appropriate to the product.

The adjustment is arranged at the swing module by regulation of two end stops. The end stops impede a pin from turning on and limit in this way the swing angle of the rotating tamp.

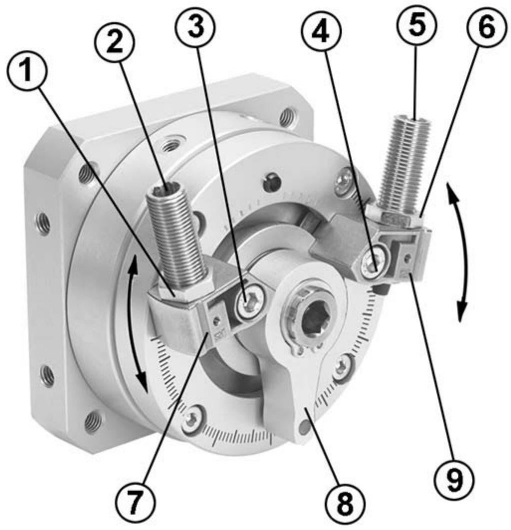


Fig. 6-4: Alpha Compact with rotating tamp

No.	Description
1	COUNTER NUT
2	*² OPTIONAL SHOCK ABSORBER
3	CLAMPING SCREW TO END STOP
4	CLAMPING SCREW TO END STOP
5	*² OPTIONAL SHOCK ABSORBER
6	COUNTER NUT
7	1. END STOP AT LH / 2. END STOP AT RH
8	PIN
9	2. END STOP AT LH / 1. END STOP AT RH

Requirements

- Labeler is powerless (power – and compressed air supply is turned off).
- Product feed is stopped.
- Labeler is aligned in position to the product.

Required equipment:

- 3mm hexagon socket wrench
- Screw driver G00

Instructions

Please adjust the rotating angle at the swing arm as follows.

Step	Procedure
1	Pull off the protective cap at the swing module.
2	Loosen the clamping screw to the 1. end position by means of a hexagon socket wrench.
3	Adjust the end position so that the tamp is horizontally in home position and that the tamp is 1-2 mm more deeply than the peeler bar.
4	Tighten again the 1. end position.
5	Loosen the 2. end position and adjust it so that the rotating tamp is parallel to the product in extended position. If the labeler is operated in Tamp On Mode, the tamp has to touch the product slightly. If the labeler is operated in Tamp Blow Mode, the tamp may not touch the product.
6	Tighten again the 2. End position.

7. Operation

Operation of the Labeler

Safety Instructions

CAUTION

Danger of drawing in caused by rotating elements.

DANGER OF DRAWING IN

Rotating elements at labeler, rewinder, web brake and rollers of the friction unit.

- Do not grip in, at or between the moving parts.
- Switch the on/off switch to position “0” during work performance at the labeler.

CAUTION

Danger caused by actively triggered movements.

DANGER OF BEING CRUSHED

Dependent on the application, the products are transferred to the labeler manually or automatically.

- Remove manually or incorrect applied labels only when you have control about the supply of products and there is no danger.

Turn on and off labeler (without power supply)

The labeler Alpha Compact is turned on and off by plugging in and out the power supply of the controller.

Turn on and off labeler with optional power supply

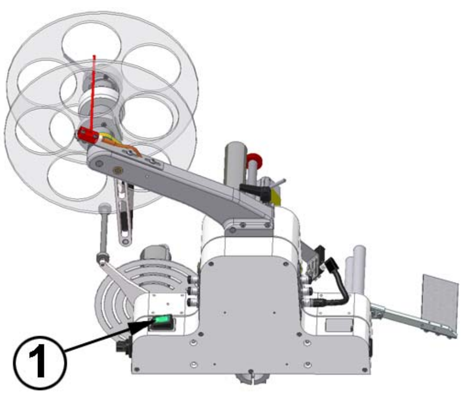


Fig. 7-1: Power supply

No.	Description
1	POWER SUPPLY

The labeler Alpha Compact is turned on and off by the *² power supply.

*² Only if labeler has the appropriate feature

Start Labeling Operation

Requirements

- The initial operation (s. chapter "Installation and Initial Operation") was finalized successfully.
- Labeler is connected with *²air pressure, power and labels are loaded.

Instructions

Please start labeling operation as follows.

Step	Procedure
1	Connect labeler with supply voltage. Afterwards there is a change to normal mode and the following text message will appear in the display.

```
ALPHA COMPACT.x     0000
STAND-BY
```

X stands for the version number of the firmware installed in the controller.

The labeler does not operate any activities in this mode. It will change to its „normal“ operating mode by pushing any button.

Step	Procedure
2	Pushing the [Start]-button activates labeling mode assumed that the starting sequence is finished and failures do not exist.

*² Only if system is equipped accordingly.

Stop Labeling Operation

Safety Instructions

NOTICE **When the system is stopped about several hours you have to take the label material off the labeler.**

The label material gets bent in the area of the deflection rollers that might lead to failures during operation mode. This label feature as well as the curving-level depend on material and can vary from label type to label type. Environmental conditions like high temperatures and high air humidity enforce this effect.

Take critical label material off the labeler before long breaks or after end of work!

Instruction

Please stop labeling operation as follows.

Step	Procedure
1	Stop product supply.
2	Switch off labeler at power switch.
3	Remove manually or faulty applied labels.
4	Take label material off labeler.

Put System out of Operation

If you put the labeler out of operation for more than 4 weeks, please perform the following steps:

Requirements

- Labeler is switched-off like in chapter “Stop Labeling Operation”.

Instruction

Please put system out of operation as follows.

Step	Procedure
1	Disconnect from power supply.
2	Transport system if necessary to its storage location like describes in chapter “Transport and Unpackaging“.
3	For dust protection cover labeler with a cotton cloth or a paper towel.
4	Air-condition storage location according to chapter „Technical Specifications“.

*² Only if system is equipped accordingly.

Operation of the HMI Panel

Fig. 7-2: HMI-Panel

The HMI Panel consists of 4 buttons, 2 control dials and 2 control-LEDs. It comprises a connection cable with a 5-pin M12 industry plug and it is connected at the labeler's side. The installation position can be freely selected and should be close to the labeler at a well accessible position. The HMI Panel is mostly used for the operation of standard labelers. Labelers with optional accessories (e.g. encoder and applicators) require for the initial operation and troubleshooting the HMI Display or the *"USB connection. A limited operation of the labelers with optional accessories is possible with the HMI Panel.

No.	Description	Specification
1	**Potentiometer m/min**	The label applying speed is adjusted with the potentiometer. In basic setting of 0-50m/min. The minimum and maximum label applying speed can be limited by means of the Display Controller by parameter (167 and 168).
2	**Potentiometer Adjust**	5 functions can be adjusted by means of the HMI Panel by parameter (021) in programming mode. 0= off / no function 1=Offset / the label position is adjusted by the poti. 2=Trigger delay / the start signal is delayed with the poti. 3=Sync pulse duration / during application cycle a synchronization signal is issued for the duration that was adjusted at the poti. 4=Sync pulse delay / the synchronization signal is issued with the delay time that was adjusted at the poti at the application cycle.
3	**LED Run**	If the green LED over the [Start]-button illuminates, "ready for labeling" is signalized.
4	**LED Status**	If the LED lights orange, the key lock for both potentiometer m/min and Adjust is turned on.

No.	Description	Specification
5	**LED Stop**	If the red LED above the [Stop]-button lights, the "labeler's standby status" is interrupted.
6	**Start**	The [**Start**]-button is used to start the labeler. The labeling operation is activated by pushing and resp. error messages are acknowledged.
	⚠ CAUTION	**Danger of drawing in by rotating elements.** - Keep away from moving parts**.**
7	**Feed**	An application cycle is arranged by pushing the [Feed] button.
	⚠ CAUTION	**Danger of drawing in by rotating elements.** - Keep away from moving parts**.**
8	**Cal**	By pushing the [Cal] button the label position is aligned in direction of the peeler bar. The labeler carries out 3 label applying procedures for calibration. Then the label's leading edge is aligned to the peeler bar.
9	**Stop**	The [**Stop**]-button is used to terminate the labeling operation. After pushing the button, the labeler changes to standby mode.
7+8	**Feed** + **Cal**	By pushing simultaneously the [Feed] and [Cal] button, a key lock for both potentiometer m/m and Adjust is turned on or off.

Operation of the HMI Display

Fig. 7-3:: Display-Controller

The HMI Display consists of a LCD-Display and 5 buttons as well as 2 control-LEDs. It comprises a connection cable with a 5-pin M12 industry plug and is connected at the labeler's back side. The installation position can be freely selected and should be close to the labeler at a well accessible position. The HMI Display provides in comparison to the HMI Panel extended display-, operation-, and service opportunities. Status- and error messages are shown by a short display message. Trained service personnel are able to put into operation additional features with the HMI Display.

No.	Term	Description
1	**Display**	Various information is indicated in the display.
2	**LED START**	If the green LED above the [Start]-button flashes, the „readiness for labeling" will be signalized.
3	**Start**	The [**Start**]-button starts the labeler. The labeling operation will be activated and maybe failure messages will be reset by pushing.
4	◀	The buttons [◀] / [▶] have 3 functions: • In combination with other buttons for selecting menus • For navigation in menus • For changing parameter values ([◀] = reduces/ [▶] = increases). The autorepeat-function of the buttons enables a quick change of high values when you hold on pushing the certain button.
	⚠CAUTION	**Danger of drawing in by rotating elements.** Via [Jog] function an applying cycle is performed. - Keep away from moving parts.

No.	Term	Description
5	Enter	The [**Enter**] key has 3 functions: • JOG-function: during labeling operation the alpha Compact can be triggered by this key. • In combination with further keys for calling up menus. • In the menus, the [Enter] key serves for changing a parameter or to confirm inputs.
6	▶	The buttons [◀] / [▶] have 3 functions: • In combination with other buttons for selecting menus • For navigation in menus • For changing parameter values ([◀] = reduces/ [▶] = increases). The autorepeat-function of the buttons enables a quick change of high values when you hold on pushing the certain button.
7	**LED Stop**	If the red LED flashes above the [Stop]-button, the „readiness for labeling“ will be interrupted.
8	Stop	The [**Stop**]-button is used to terminate the labeling operation. After pushing the button, the labeler changes to standby mode.
3+5	◀ + ▶	By pushing the [◀] + [▶] and [Cal] button simultaneously the label position is aligned in direction of the peeler bar. The labeler carries out 3 label applying procedures for calibration. Then the label's leading edge is aligned to the peeler bar.

Program-Menus

Safety Instructions

NOTICE

The PROGRAMMING MODE is only allowed to be used by qualified persons!

- A false programming may cause failures!

The display controller of the Alpha Compact includes three menu modes which can be reached and left via key combination. Push the following buttons together at the same time in stand-by status to get to the desired menu:

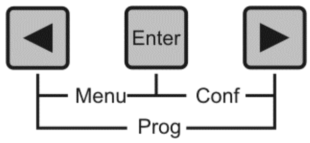


Fig. 7-4: Buttons below Display

[◀] + [**Enter**] = **FUNCTION-MENU**

[**Enter**] + [▶] = **CONFIGURATION-MODE**

[◀] + [▶] = **PROGRAMMING-MODE**
(Only for qualified personnel, further information in the service manual)

Here is a display example for further explanations:

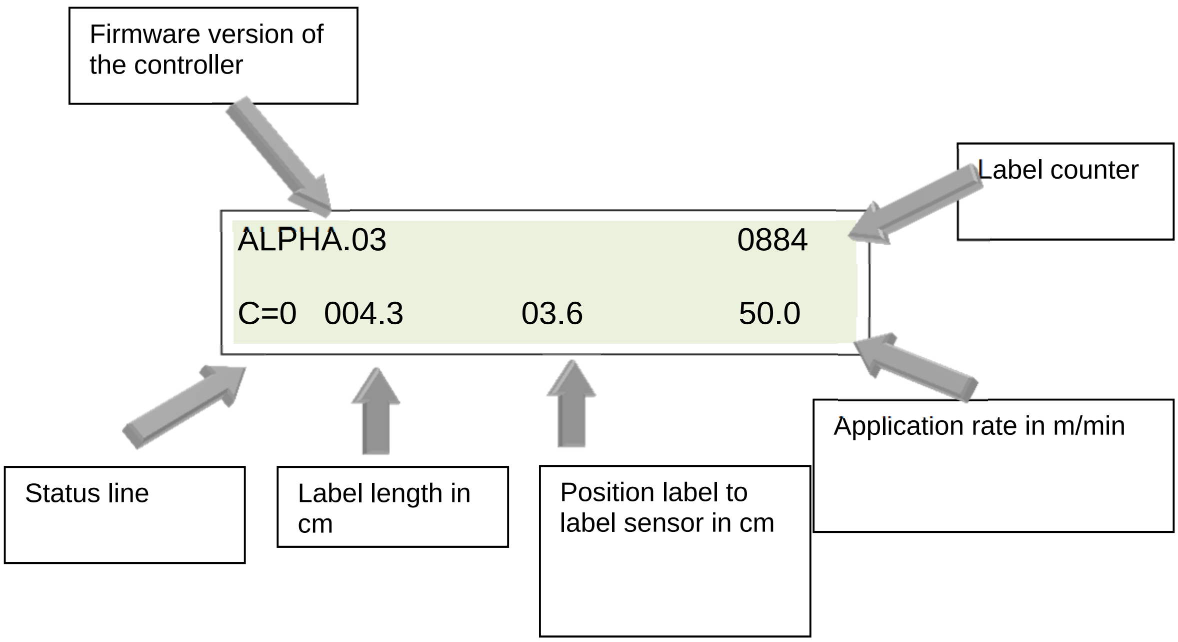


Fig. 7-5: Display Explanation

Failure Reset

By pushing [Start] or [Stop] button the current error messages are reset assuming that the reason for the error was corrected.

Indication of Firmware Version

Here the current firmware-version (Firmware = operating system) of the controller is indicated (firmware-history on page 113).

Current Configuration Dataset

The Alpha Compact offers the opportunity to store up to 4 different sets of parameters. Each dataset can be used for an individual parameterization of the labeler (e.g. for different production conditions).

Label Counter

The 5-digit figure shows the number of the completely performed applying cycles during an operating phase. The counter is reset automatically to "00000" after switching on the labeler. If a 4-digit negative figure is shown, the counter will work in "countdown" mode.

Status Line

The status line informs you about the current status of the labeler and its activities.

If a 3-digit “information figure” is indicated on the right side of the status line, it will give you additional information and will be explained in detail in the chapter troubleshooting.

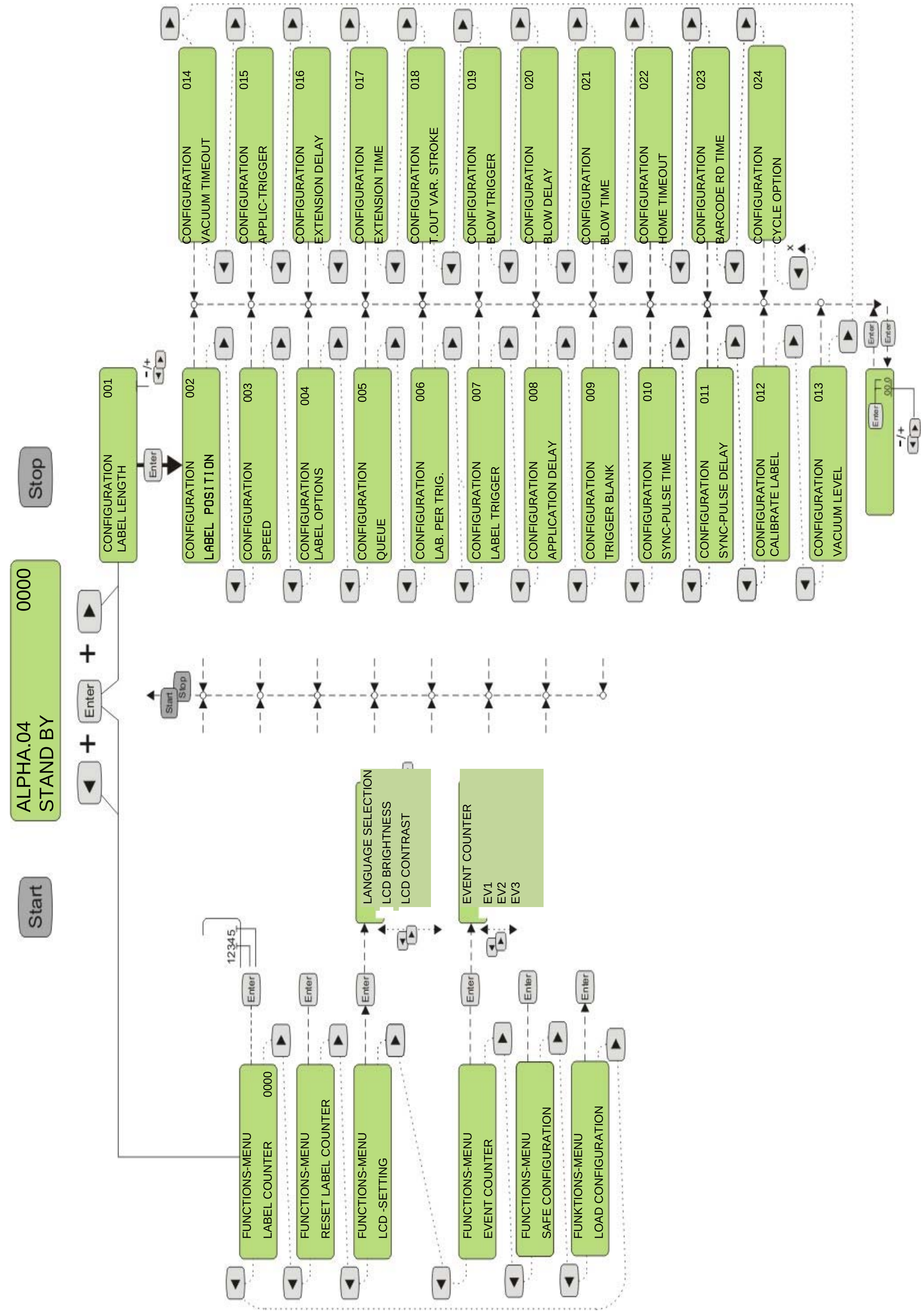
STAND-BY Mode

```
ALPHA.03          0000
STAND-BY
```

The message "Stand-by" is indicated after a manual interruption of the labeling operation by pushing the [Stop]-button. A menu of the labeler`s controller can only be selected based on stand-by-mode. (The control-LED GREEN is off and RED flashes)

Standard labelers that are supplied ex-stock are preset in English menu language. In function menu by LCD –SETTING (DISPLAY SETTING) see **p. 87** the menu language can be changed to e.g. German.

Menu-Diagram



FUNCTION-MENU

You can reach the function menu by pushing [◀] and [Enter] at the same time assuming that the labeler is in stand-by (s.a.). This menu includes some basic settings of the labeler with the following submenus:

FUNCTION-MENU

- LABEL COUNTER
- RESET COUNTER
- LCD SETTINGS
- LANGUAGE SELECTION (Option)
- EVENT COUNTER
- SAVE CONFIGURATION
- LOAD CONFIGURATION

LABEL COUNTER

Here you have the opportunity to determine the label counter a certain value between 0-25599. You activate the countdown function by an input value larger “00000”. That means the input value will be counted down during each applying cycle in the labeling mode. The display indicates a “minus” forward the counter value. The counter value corresponds to the remaining applying cycles. The labeler will stop automatically if the final value “-0000” is reached.

FUNCTION-MENU
LABEL COUNTER 01234

- Push the * buttons [◀] or [▶] to reach the next parameter.
- Push [Enter] to edit the front digits or push again [Enter] to change to the back digits.
- Push the * buttons [◀] or [▶] to increase or reduce the edited parameter value (cursor flashes).
- Push [Start] or [Stop] to leave the parameter settings.

*Fast forward-function when pushing the button [◀] or [▶].

RESET LABEL COUNTER

The counter reading of the label counter will be reset by this function.

```
FUNCTION-MENU

RESET LABEL COUNTER
```

- Push the * buttons [◀] or [▶] to reach the next parameter.
- Push [Enter] to reset the counter to "00000".
- Push [Start] or [Stop] to leave the parameter settings.

LCD -SETTING

This submenu includes display settings. Here you can change contrast and lightness.

```
FUNCTION-MENU
LCD-SETTING
```

- LANGUAGE SELECTION >>	Selects the certain language.
- LCD-LIGHTENING >>	Lightness of the backlight Value between 001-040.
- LCD CONTRAST >>	Contrast setting between 001-100

```
LCD-SETTING
LANGUAGE SELECTION
LCD-LIGHTNESS:  015
LCD-CONTRAST:   030
```

- Push the * buttons [◀] or [▶] to reach the next display setting.
- Push [Enter] to change the parameter.
- Push the * buttons [◀] or [▶] to increase or reduce the edited parameter value (cursor flashes).
- Push [Enter] to store the value.
- Push [Start] or [Stop] to leave the parameter settings.

*Fast forward-function when pushing the button [◀] or [▶].

EVENT COUNTER

Using this function a number of different event counters can be called up. These provide useful statistic information about the events which are counted automatically during labeling operation.

Event	Counter Information
E00	Number of all stopped applying cycles without failures
E01 until E22	Number of failures of the certain error code
EV1	*Number of printed labels
EV2	*Number of all extension movements of the cylinder.
EV3	*Number of blow off events
EV4	*Number of all communication failures (incl. autom. restored failures)
EV5	*Maintenance counter increases by 1 after the completion of 256 applying cycles

*All counter values are stored in the memory of the labeler. The time at which to store the counter reading happens during a failure or in stand-by mode. This means if the system is switched off in "READY"-Mode, the counter reading of EV1-EV5 can differ extremely from the actual counted values.

FUNCTION-MENU
EVENT-COUNTER

- Push the *buttons [◀] or [▶] to change to the next parameter.
- Push [Enter] to get to the event counters.

E00=00000
E01=00000
.
.
E22=00000

5-digit counter,
maximum value = 65535
(afterwards begin with 00000)

EV5=00000
EV1=00000000
.
.
EV4=00000000

8-digit counter
Maximum value: 99999999
(afterwards begin with 00000000)

- Push the buttons [◀] or [▶] to get to the next event counter.
- Push [Start] or [Stop] to leave the parameter settings.

*Fast forward-function when pushing the button [◀] or [▶].

SAVE CONFIGURATION

The Alpha Compact’s controller is able to store up to 9 different configurations. This feature provides the opportunity to the user for a fast product change. Thus all topical settings (also the settings from configuration menu) are stored e. g. speed, Encoder turned on or off, application delay:

```
FUNCTION-MENU
SAVE CONFIGURATION
```

- Push the *buttons [◀] or [▶] to get to the next parameter.
- Push [Enter] to select ”save configuration“. A number will flash in the display. This number is the number of data set in which the configuration should be saved.
- Push the buttons [◀] or [▶] to select the required data set.
- Push [Enter] to save the configuration in the selected data set.
- Push [Start] or [Stop] to leave the parameter setting.

LOAD CONFIGURATION

If a configuration is saved, it can be loaded and activated.

```
FUNCTION-MENU
LOAD CONFIGURATION
```

- Push the *buttons [◀] or [▶] to get to the next parameter.
- Push [Enter] to load the required data set in the controller again. A number will flash in the display. This number is the number of the data set.
- Push the buttons [◀] or [▶] to select the required data set.
- Push [Enter] to load the configuration of the selected data set in the controller.
- Push [Start] or [Stop] to leave the parameter setting.

CONFIGURATION-Menu

The configuration menu can be reached by pushing the buttons [Enter] and [▶] at the same time assuming that the labeler is in stand-by.

This menu includes parameter settings for the whole labeler configuration which determine the sequence of events during an applying cycle.

On page 100 you have the opportunity to enter your parameters in a list.

CONFIGURATIONS-Address

The display shows in the first line behind "CONFIGURATION" a number with 3 digits (e.g. "001"). Thus the number represents the currently selected address. In line 2 the respectively selected parameter is noted.

```
CONFIGURATION          001
LABEL LENGTH CM:
```

PARAMETER IN CONFIGURATIONS-MENU	ADDRESS
LABEL LENGTH CM:	001
LABEL POSITION CM:	002
SPEED M/MIN	003
LABEL OPTIONS	004
WAITING QUEUE	005
LABELS PER TRIG.	006
LABEL TRIGGER	007
PROD. DELAY	008
ING. TRIGGER	009
SYNC-PULSE TIME	010
SYNC-PULSE DELAY	011
CALIBRATE LABEL	012
VACUUM LEVEL:	013
VACUUM TIMEOUT:	014
APPLIC-TRIGGER:	015
EXTENSION DELAY	016
EXTENSION TIME	017
PROXIMITY T. OUT	018
BLOW TRIGGER:	019
BLOW DELAY	020
BLOW TIME	021
HOME TIMEOUT:	022
BARCODE READING TIME	023
CYCLE OPTION:	024

PASSWORD: 000

If a configuration parameter is called up for editing by [Enter], you will have to enter a password in order to grant access only to authorized persons.

The code is “123”, the access is valid for all parameters, but only as long as you do not push 2 times the [Stop]-button or switch off and on the labeler.

ALPHA.04	00000
PASSWORD:	000

- Push [Enter] to edit the configuration parameter (in case of password query see page **91)**.
- Push the *buttons [◀] or [▶] to increase or reduce the value.
- Push [Enter] to confirm the password entry.
  - If password is correct you will return to configuration-parameter.
  - If password is incorrect, "PASSWORD INCORRECT E15" will appear.
- Push [Start] or [Stop] to leave the parameter setting.

* Fast forward-function when pushing the button [◀] or [▶].

001 LABEL LENGTH (LABEL LEN.)

Herewith the label length (label length with a gap) is manually adjusted. There is the possibility of automatic label calibration.

CONFIGURATION	001
LABEL LENGTH	

- Push [Enter] to edit the configuration parameter (in case of password query see page **91)**.
- Push the *buttons [◀] or [▶] to increase or reduce the value.
- Push [Enter] to confirm the entry.
- Push [Start] or [Stop] to leave the parameter setting.

* Fast forward-function when pushing the button [◀] or [▶].

002 LABEL POSITION (LABEL POS.)

Herewith the label position to the peeler bar is adjusted manually. There is the possibility of automatic label calibration.

CONFIGURATION	02
LABEL POSITION cm:	11,8

- Push [Enter] to edit the configuration parameter (in case of password query see page **91)**.
- Push the *buttons [◀] or [▶] to increase or reduce the value.
- Push [Enter] to confirm the entry.
- Push [Start] or [Stop] to leave the parameter setting.

* Fast forward-function when pushing the button [◀] or [▶].

003 SPEED (SPEED)

Here the application rate is adjusted. This speed should correspond to the speed at which the product is transferred to the labeler. By using a rotating shaft encoder there is the possibility that the controller identifies automatically the product speed and also reacts on varying product speeds.

CONFIGURATION	03
SPEED m/min:	0.50

- Push [Enter] to edit the configuration parameter (in case of password query see page **91)**.
- Push the *buttons [◀] or [▶] to increase or reduce the value.
- Push [Enter] to confirm the entry.
- Push [Start] or [Stop] to leave the parameter setting.

Detailed information on application rate

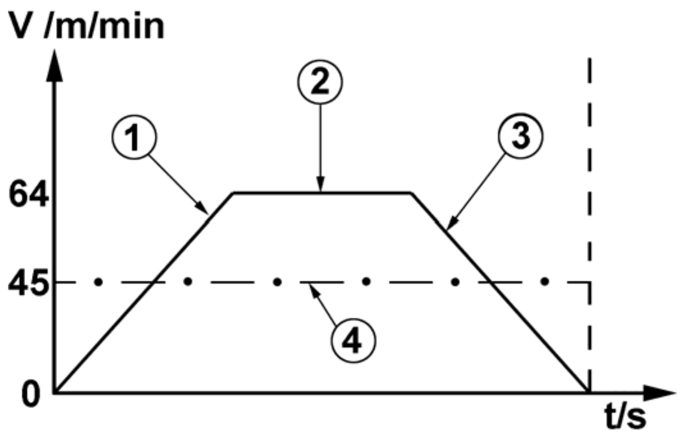


Fig. 7-6 Label Feed

No.	Description
1	START RAMP
2	LABEL FEED SPEED
3	STOP RAMP
4	AVERAGE SPEED

Acceleration (start ramp) and delay (stop ramp) of label feed speed are generally linear. The average speed results from acceleration, period of feed and delay. Let`s assume that a 500 mm long label with a gap of 3 mm should be applied at a feed speed of 64 m/min, it corresponds to an average of 45m/min (because the ramps take a formidable part of the label`s total length).

The start-/stop-ramps can be adjusted in 4 stages by the parameter 04 LABEL OPTIONS (LABEL OPTIONS).

At really smoothly adjusted acceleration (start ramp) and delay (stop ramp), the adjusted feed speed cannot be achieved, particularly with short labels.

If a shaft encoder should be used, the parameter 04 LABEL OPTIONS (LABEL OPTIONS) has to be adjusted to +64.

Tables on acceleration ramps for 2 power supply version:

Clock rates of the 3 power supply version can be taken from the performance calculator.

	2G		3G		4G		5G	
m/min	**mm**	**ms²**	**mm**	**ms²**	**mm**	**ms²**	**mm**	**ms²**
10	0,8	7	0,6	5	0,5	3	0,3	2
20	2,9	15	2,0	10	1,5	7	1,2	6
30	6,6	24	4,4	15	3,2	11	2,6	9
40	11,6	32	7,7	212	5,8	15	4,6	12
50	18,2	41	12,1	27	8,9	19	7,2	15

004 LABEL OPTIONS

Herewith the acceleration ramp for label application is adjusted. You can also select further functions. The functions are combined with each other with the help of arithmetic sum or are activated separately.

CONFIGURATION	004
LABEL OPTIONS:	094

- Push [Enter] to edit the configuration parameter (in case of password query see page **91)**.
- Push the *buttons [◀] or [▶] to increase or reduce the value.
- Push [Enter] to confirm the entry.
- Push [Start] or [Stop] to leave the parameter setting.

Parameter	Function
000	The acceleration ramp 2G is selected.
+1	The acceleration ramp 3G is selected.
+2	The acceleration ramp 4G is selected.
+3	The acceleration ramp 5G is selected.
+4	Doubles the braking ramp.
+32	*selects the pneumatic peeler blade.
+64	Selects the optional connected rotary shaft encoder.
+128	Selects the trailing edge as reference edge.

*If the parameter +32 is choosen, triggering occurs at the abatement of the pneumatic peeler blade. After that the label is dispensed. The time of dispensing the label can be setted via parameter PROD.VERZOEGERUNG (TRIGGER DELAY). Default parameter is 4+32=36. For some application it is better when label dispensing and abatement of pneumatic peeler blade occur at the same time. In this case parameter +32 is not choosen.

The duration of abatement of the pneumatic peeler blade has to be setted with parameter SYNC-IMPULS-ZEIT (SYNC PULSE TIME).

05 LABEL QUEUE (LABEL QUEUE)

Herewith the number of labels between peeler bar and label sensor is adjusted manually. There is the possibility of automatic label calibration.

CONFIGURATION	005
LABEL QUEUE	002

- Push [Enter] to edit the configuration parameter (in case of password query see page **91)**.
- Push the *buttons [◀] or [▶] to increase or reduce the value.
- Push [Enter] to confirm the entry.
- Push [Start] or [Stop] to leave the parameter setting.

006 LABELS BURST (LABELS BURST)

Herewith the number of label application per start signal (trigger signal) is adjusted. If the parameter is set on 000, 1 label per start signal will be applied.

CONFIGURATION	006
LABELS BURST	001

- Push [Enter] to edit the configuration parameter (in case of password query see page **91)**.
- Push the *buttons [◀] or [▶] to increase or reduce the value.
- Push [Enter] to confirm the entry.
- Push [Start] or [Stop] to leave the parameter setting.

007 LABEL TRIGGER (LABEL TRIGGER)

Herewith it is determined which sensor input and what kind of signal generates the start signal for labeling. The functions are combined with each other with the help of arithmetic sum or are activated separately.

CONFIGURATION	007
LABEL TRIGGER	001

- Push [Enter] to edit the configuration parameter (in case of password query see page **91)**.
- Push the *buttons [◀] or [▶] to increase or reduce the value.
- Push [Enter] to confirm the entry.
- Push [Start] or [Stop] to leave the parameter setting.

Parameter	Function
000	As soon as the labeler is ready, the application starts.
001	The first (primary) product sensor input is selected as start sensor.
002	The second (secondary) product sensor input is selected as start sensor.
+4	As start signal not a switching edge is used but an input level. As long as an input level is available, the application cycle is repeated. This enables e. g. the multiple labeling of products with different lengths.
+8	A start signal is only accepted after terminated labeling applying cycle for an again label application. If the label applying cycle is not yet terminated, an early start signal leads to error message E07. This function cannot be combined with param-eter +004.
+16	Selects leading edge as starting signal.
+32	Enables the connection of a NPN sensor as start signal-sensors. (Standard is a PNP sensor).

008 TRIGGER DELAY (TRIGGER DELAY)

Herewith a delay is adjusted between trigger and label application. The delay can be between 0 up to app. 255 mm. The length of the pulse corresponding to each entered value x 1 mm. Because of the acceleration of the motor is a linear adjustment not possible (hence app. 1mm).This feature can be used to change the label position without mechanical adjustment of the start sensor.

CONFIGURATION	008
TRIGGER DELAY:	001

- Push [Enter] to edit the configuration parameter (in case of password query see page **91)**.
- Push the *buttons [◀] or [▶] to increase or reduce the value.
- Push [Enter] to confirm the entry.
- Push [Start] or [Stop] to leave the parameter setting.

009 TRIGGER BLANK

With parameter 009 IGN TRIGGER , a start trigger debouncing can be adjusted, e.g. at products with foil packaging.

If in parameter „006 LABELS PER TRIG“. More than one label per triggering was selected, with parameter 009 IGN TRIGGER the distance between the single label applications is adjusted.

If a shaft encoder is connected, the adjustment is arranged in millimeters. If there is no shaft encoder connected, the adjustment is arranged in milliseconds.

CONFIGURATION	009
TRIGGER BLANK	000

- Push [Enter] to edit the configuration parameter (in case of password query see page **91)**.
- Push the *buttons [◀] or [▶] to increase or reduce the value.
- Push [Enter] to confirm the input.
- Push [Start] or [Stop] to leave the parameter setting.

*Fast forward-function when pushing the button [◀] or [▶].

010 SYNC PULSE TIME (SYNC PULSE TIME)

Herewith the duration of the synchronization output signal is determined. This signal serves for triggering further peripheries like rotating tamp, printer etc. The signal is activated for the adjusted time at the end of an applying cycle. The time can be adjusted from 1 up to 255ms. The pulse time is corresponding to each entered value x 10ms. If the value is set on “000” and the SYNC PULSE DELAY as well on zero. The SYNC signal will be activated with the beginning of the applying cycle and deactivated at the end of the applying cycle.

CONFIGURATION	010
SYNC PULSE TIME:	000

- Push [Enter] to edit the configuration parameter (in case of password query see page **91)**.
- Push the *buttons [◀] or [▶] to increase or reduce the value.
- Push [Enter] to confirm the entry.
- Push [Start] or [Stop] to leave the parameter setting.

011 SYNC PULSE DELAY (SYNC PULSE DELAY)

With this option, the duration of a delay of the synchronization output signal after the labeling procedure is determined. It is used to calibrate the triggering of further peripherals like rotating tamp, printer aso. The duration can be adjusted from 0 until 255ms. The length of the pulse corresponds tot he entered value x 10 ms. If the SYNC-PULSE TIME is set to “000“, this function can be used to evaluate „Labeling cycle in progress“.

CONFIGURATION	011
SYNC PULSE DELAY	000

- Push [Enter] to edit the configuration parameter (in case of password query see page **91)**.
- Push the *buttons [◀] or [▶] to increase or reduce the value.
- Push [Enter] to confirm the entry.
- Push [Start] or [Stop] to leave the parameter setting.

012 CALIBRATE LABEL (CALIBRATE LABEL)

This function is used to perform an automatic label calibration. An application of several labels happens. Thereby parameters like label position and label length will be determined. If a calibration is successfully finished, the values will be indicated in the second line. If the calibration is not finished successfully, “CALIBRATE LABEL” remains in the second line and the process will have to be repeated.

CONFIGURATION	012
CALIBRATE LABEL	

- Push [Enter] to start calibration.
- Push [Start] or [Stop] to leave the parameter setting.

013 VACUUM LEVEL

This parameter determines the threshold for label presence detection on tamp. The high-sensitive detector recognizes least leakages of pressure what results into a practicable range of adjustment of 0-10 units. If the value "003" is entered a sufficient vacuum check is realized. Values beyond, belong to the sensitive or high-sensitive measure range. If the value is adjusted to 000, there is no vacuum check active.

CONFIGURATION	013
VACUUM LEVEL	000

- Push [Enter] to edit the configuration parameter (in case of password query see page **91)**.
  - Push the *buttons [◀] or [▶] to increase or reduce the parameter value. (unit = value x 1/33 bar, in case of value = "0" the function is switched off.).
  - Push [Enter] to save the value.
- Push the buttons [◀] or [▶] to get to the next configuration parameter.
- Push [Start] or [Stop] to leave the parameter setting.

**Fast forward-function when pushing the button [◀] or [▶].

014 VACUUM TIMEOUT: 200

This parameter determines the maximum time delay the Alpha Compact will try to detect label presence on tamp by sensing the vacuum in the tamp before generating an error message. This means the achievable vacuum has to be complete within the adjusted time, otherwise an error occurs. Normally the vacuum is controlled <u>after</u> printing procedure. For particular applications, the value can be set to “000” whereby the examination of the vacuum is started together with printing.

CONFIGURATION	014
VACUUM TIMEOUT	200

- Push [Enter] to edit the configuration parameter (in case of password query see page **91**).
  - Push the *buttons [◀] or [▶] to increase or reduce the parameter value. (unit = value x 10ms, value = "0" abrupt vacuum testing, in case of value = "255" the labeler waits indefinitely until the vacuum was set up (200 = 2 seconds = default).
  - Push [Enter] to save the value.
- Push the buttons [◀] or [▶] to get to the next configuration parameter.
- Push [Start] or [Stop] to leave the parameter setting.

**Fast forward-function when pushing the button [◀] or [▶].

015 APPLIC-TRIGGER : 000

Herewith the trigger signal for the extension of the applicator is parameterized. After the label was detected at the stamps, the date is in the labeling cycle at which the trigger is expected.

Parameter Value	Connection	Function
00		Immediate extension of applicator.
01 02 03 04	Product Sensor 1 (J14) Product Sensor 2 (J9) Aux.-input at I/O-interface: J16 Applicator J3	Edge triggering (rising edge): Triggering in case of a rising edge, means the change from "low" to "high". Only in case of indication "Expect. Trig." A trigger is accepted. Outside this phase all triggers are ignored.
05	Product Sensor 1 (J14)	Level triggering (rising edge): Triggering in case of "high"-signal. Repetition of applying cycle as long as signal is active (high).
29 - 32	No acceptable parameters.	

```
CONFIGURATION                015
APPLIK. TRIGGER          000
```

- Push [Enter] to edit the configuration parameter (in case of password query see page **91**).
  - Push the *buttons [◀] or [▶] to increase or reduce the parameter value (valid range: 000-032).
  - Push [Enter] to save the value.
- Push the buttons [◀] or [▶] to get to the next configuration parameter.
- Push [Start] or [Stop] to leave the parameter setting.

*Fast forward-function when pushing the button [◀] or [▶].

016 EXTENSION DELAY

This parameter enables the programming of a delay between trigger signal and actual tamp movement. The delay starts with triggering (external or autom.). The EXTENS: DELAY will be ignored, if the parameter APPLICATOR TRIGGER has the value “000” (cp. page **91.)**

CONFIGURATION	016
EXTENS. DELAY	000

- Push [Enter] to edit the configuration parameter (in case of password query see page 91).
  - Push the *buttons [◀] or [▶] to increase or reduce the parameter value. (unit = value x 10ms, value "000" = no delay).
  - Push [Enter] to save the value.
- Push the buttons [◀] or [▶] to get to the next configuration parameter.
- Push [Start] or [Stop] to leave the parameter setting.

017 EXTENSION TIME

The „EXTENSION TIME“ defines the time that passes from the beginning of the stroke movement until the beginning of blow-action. You set the labeler in Blow-On mode via the value “000”. The label will be blown off directly without movement of the applicator.

The dimensioning of the adjusted time (unit = value x 10ms) should correspond to the actual time for the extension of the whole stroke. If the specified value is exceeded because of another parameter configuration, the label will be blown off immediately onto the product, unless the labeler waits of a trigger signal (e.g. “BLOW TRIGGER” or from the sensor “variable stroke”). When using the option “variable stroke” normally the EXTENSION TIME is set on a low value (e.g. “002”) to continue with the variable range of the stroke after a constant extension length.

CONFIGURATION	017
EXTENSION TIME	080

*Fast forward-function when pushing the button [◀] or [▶].

- Push [Enter] to edit the configuration parameter (in case of password query see page 91).
  - Push the *buttons [◀] or [▶] to increase or reduce the parameter value. (unit = value x 10ms, value "000" = Blow-On mode).
  - Push [Enter] to save the value.
- Push the buttons [◀] or [▶] to get to the next configuration parameter.
- Push [Start] or [Stop] to leave the parameter setting.

018 TIME OUT VARIABLE STROKE

Any parameter value higher “000” activates the function “variable stroke” which requires an optional sensor at the tamp pad. The value you enter defines the maximum period the applicator waits during extension, in which the variable stroke sensor has to detect a product. Otherwise an error message will result. The waiting time will be displayed and counted down.

In case of a parameter value of „255“ it will be waited indefinitely for a triggering of the proximity sensor (no time is displayed).

CONFIGURATION	018
T.OUT VAR. STROKE	000

- Push [Enter] to edit the configuration parameter (in case of password query see page **91).**
  - Push the *buttons [◀] or [▶] to increase or reduce the parameter value. (unit = value x 100ms, value has to be "000" when the option „variable stroke“ is not installed, in case of value "255" = waits indefinitely).
  - Push [Enter] to save the value.
- Push the buttons [◀] or [▶] to get to the next configuration parameter.
- Push [Start] or [Stop] to leave the parameter setting.

*Fast forward-function when pushing the button [◀] or [▶].

019 BLOW TRIGGER

Herewith the trigger signal for the blow-off of a label is adjusted. In case of value "000" the label will be blown off without delay or rather automatically. Higher values stop the applying cycle at this point and the applicator waits for the pre-defined trigger signal.

Depending on the configuration additionally the trigger can be used for the control of the external stop function or rather for starting the labeling operation.

If the Alpha Compact operates with a variable stroke, the value has to be "000" according to factory setting.

Parameter Value	Connection	Function
00		Automatic blow-off of the label.
01 02 03 04	Product Sensor 1 (J14) Product Sensor 2 (J9) Aux.-input at I/O-interface: J16 Applicator J3	Edge triggering (rising edge): Triggering in case of a rising edge, means the change from "low" to "high". Only in case of indication "Expect. Trig." A trigger is accepted. Outside this phase all triggers are ignored.
05	Product Sensor 1 (J14)	Level triggering (rising edge): Triggering in case of "high"-signal. Repetition of applying cycle as long as signal is active (high).
29 - 32	No acceptable parameters.	

```
CONFIGURATION          019
BLOW TRIGGER           000
```

- Push [Enter] to edit the configuration parameter (in case of password query see page **91).**
  - Push the *buttons [◀] or [▶] to increase or reduce the parameter value. (valid range: 000-032).
  - Push [Enter] to save the value.
- Push the buttons [◀] or [▶] to get to the next configuration parameter.
- Push [Start] or [Stop] to leave the parameter setting.

*Fast forward-function when pushing the button [◀] or [▶].

020 BLOW DELAY: 000

This configuration parameter delays the blow-off by the adjusted value. The adjustment will be ignored when the previous parameter „BLOW TRIGGER“ is set on “000“.

CONFIGURATION	020
BLOW DELAY.	000

- Push [Enter] to edit the configuration parameter (in case of password query see page **91).**
    - Push the *buttons [◀] or [▶] to increase or reduce the parameter value. (unit = value x 10ms, value "000" = no delay).
    - Push [Enter] to save the value.
- Push the buttons [◀] or [▶] to get to the next configuration parameter.
- Push [Start] or [Stop] to leave the parameter setting.

021 BLOW TIME

The blow time and thus the duration of the blow-off valve`s activation for the transfer of the label on to the product is determined. The allowed range is between 010 and 255 (value x 1ms).

CONFIGURATION	021
BLOW TIME	000

- Push [Enter] to edit the configuration parameter (in case of password query see page **91**).
    - Push the *buttons [◀] or [▶] to increase or reduce the parameter value. (unit = value x 1ms, value "000" Tamp-On mode).
    - Push [Enter] to save the value.
- Push the buttons [◀] or [▶] to get to the next configuration parameter.
- Push [Start] or [Stop] to leave the parameter setting.

*Fast forward-function when pushing the button [◀] or [▶].

022 HOME TIMEOUT

This adjustment determines the maximum time the applicator waits for the signal of the home position sensor during retract movement before an error message is indicated. This is also valid after switching on the labeler in case the tamp does not move to home position because e.g. air pressure has not yet been connected.

CONFIGURATION	022
HOME TIMEOUT	000

- Push [Enter] to edit the configuration parameter (in case of password query see page **91).**
    - Push the *buttons [◀] or [▶] to increase or reduce the parameter value (unit = value x 100ms, value "000" = no query of home position, value "255" = waits indefinitely for home position signal).
    - Push [Enter] to save the value.
- Push the buttons [◀] or [▶] to get to the next configuration parameter.
- Push [Start] or [Stop] to leave the parameter setting.

023 BARCODE READ TIME

This adjustment determines the maximum time the applicator waits for the signal of the BARCODE reader before an error message is indicated.

CONFIGURATION	023
BARCODE READ TIME	000

- Push [Enter] to edit the configuration parameter (in case of password query see page **91).**
    - Push the *buttons [◀] or [▶] to increase or reduce the parameter value (unit = value x 100ms, value "000" = no query of home position, value "255" = waits indefinitely for home position signal).
    - Push [Enter] to save the value.
- Push the buttons [◀] or [▶] to get to the next configuration parameter.
- Push [Start] or [Stop] to leave the parameter setting.

*Fast forward-function when pushing the button [◀] or [▶].

024 CYCLE OPTION: 000

Pre-defined program processes can be selected here. The processes can be combined or activated separately with the help of arithmetic sums:

+1 deactivates the timeout "variable stroke“ and generates an error if the label is lost during tamp movement (vacuum loss);

+2 delays the tamp retract at the end of a blow-off sequence (normally the tamp moves back concurrently with the blow-off valve`s activation).

+4 deactivates the timeout "variable stroke“ and ignores vacuum loss during tamp movement.

+8 activates corner wrap application function for systems with 90°-180° rotating tamp (option).

+ 16 activates the vacuum or rather “label on pad“ query after label transfer (during tamp retraction) in case of Tamp-On mode.

+32 activates the function for the control of RFID reject unit.

+64 deactivates the error “timeout label“. (deaktiviert den Fehler "Timeout Etikett". (useful function to correct errors of vacuum, to switch-off the vacuum check).

+128 deactivates the timeout-error in case of variable stroke.

CONFIGURATION	024
CYCLE OPTION	000

- Push [Enter] to edit the configuration parameter (in case of password query see page **91).**
  - Push the *buttons [◀] or [▶] to increase or reduce the parameter value (value: "000" = no program sequence selected, "001" = 1st function; "002" = 2nd function, "003" = 1st and 2nd function…).
  - Push [Enter] to save the value.
- Push the buttons [◀] or [▶] to get to the next configuration parameter.
- Push [Start] or [Stop] to leave the parameter setting.

*Fast forward-function when pushing the button [◀] or [▶].

PROGRAMMING

NOTICE

Wrong parameters can lead to bugs and improper functions, also possibly leading to mechanical damage.
This applies also in changes of reserved or undocumented parameters as well as entering values outside the valid ranges.

– The parameterization should only be edited by authorized and trained personnel!

The PROGRAMMING mode allows the consideration of all 256 service parameters and enables the access to a further sub-menu. For a change of the values, a 5-digit password has to be entered.

After correct entry of the password, the access to several system parameters is permitted and presumes that the user provides specific knowledge and has read and understood all parts of this chapter.

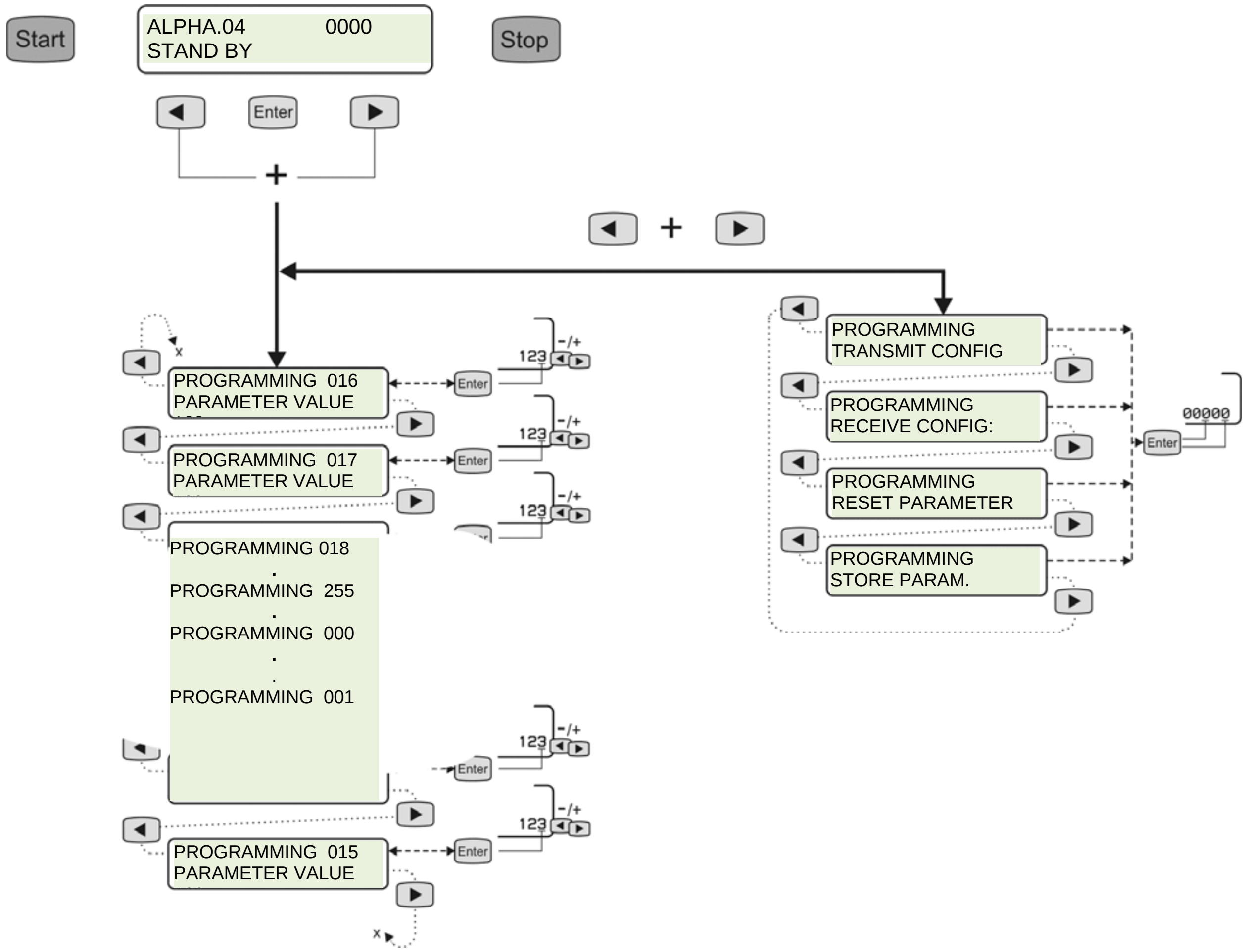
Most of these parameters remain unchanged in their factory settings (shortly "default"). But few of them may need to be adapted to specific requirements (probably just once during installation). A further sub menu allows to transmit/receive and to save/repair all system parameters. In the following an overview of available items (more detailed in the chart on page 108).

PROGRAMMING…..

PARAMETER VALUE: xxx	RECEIVE CONFIG.
	TRANSMIT CONFIG.
	RESET PARAMETER
	STORE PARAM.

The display shows in line 1, after „PROGRAMMING“ the continuous parameter number (e.g. “016”) of the collective 256 parameters. The value (resp. the content) of the parameter is in line 2 on the righthand side.

Overview chart PROGRAMMING



Call Up, Navigate and Edit Programming

Navigation and processing are the same for all 256 parameters are explained here representatively once.

Requirements

- The HMI-display is in Stand-by-operation.

Instructions

Access to PROGRAMMING is granted as follows:

Step	Procedure
1	Access to PROGRAMMING is granted (from stand-by mode) by pushing the buttons [◀] and [▶] simultaneously
2	Following possibilities are there:

```
PROGRAMMING   016
PARAMETER VALUE:   000
```

- Push [Enter] to edit parameter (at password request s. page **91**)
  - Push then the *buttons [◀] or [▶] to reduce or increase the parameter value
  - To save the value, press [Enter].
- Push the buttons [◀] or [▶] to get to the next parameter
- To exit PROGRAMMING, push [Stop].

Provided that you are in PROGRAMMING mode, you will receive access to a sub menu with 4 more functions concerning the complete configuration of the Alpha Compact by pushing the buttons [◀] and [▶].

```
PROGRAMMING
TRANSMIT CONFIG.
```

Further information on navigation from page **109**.

* Fast forward-function when pushing the button [◀] or [▶].

TRANSMIT PARAMETERS

This function allows the Alpha Compact to send its configuration parameters completely to an external PC linked to the USB port.

By pushing the [Enter] button, the data will be transmitted to USB port.

To clone the configuration of a master unit, the output may be stored as a text file and re-transmitted to other Alpha Compact machines. The text files can be edited on a PC and retransmitted to the labeler and thus an external service programming is possible.

PROGRAMMIUNG
TRANSMIT PARAM.

- Press either [◀] or [▶] key* to move to the next parameter
- Press [Enter] to transmit the configuration parameters.
- Press the [◀] and [▶] buttons simultaneously to exit this sub menu.
- To exit PROGRAMMING press the [Start] or [Stop] key.

RECEIVE PARAMETERS

NOTICE **Improper data received through this function may lock-up the machine or cause unpredictable behaviour! (s. "RESET PARAMETERS)**

This function sets the machine in a waiting condition to receive configuration parameters from an external PC linked to the USB port.

The function is activated by pushing the [Enter] button. A 5-digit password has to be entered. The receiving mode is terminated automatically by transmission of the last data file. It is also possible to terminate the receiving mode by pushing the [Stop]-button.

PROGRAMMING
RECEIVE PARAM.

- Press either [◀] or [▶] key* to move to the next parameter
- Press [Enter] to receive the configuration parameters (if password is requested, see page **91**). [Stop] interrupts receiving mode
- Press the [◀] and [▶] buttons simultaneously to exit this sub menu
- To exit PROGRAMMING press the [Start] or [Stop] key

RESET PARAMETER

This function allows to reset the operating parameters to default values that are initially pre-defined from factory, but that may also be later customized according to specific application requirements (s. "STORE PARAMETERS" on page **111**). It is useful to restore a known and safe configuration in case of improper parameter modifications.

To execute the function, the 5-digits password is required (default value: 01234). Only parameters 000 to 188 are restored, the others (event counter and checksum) remain unchanged. After execution, the machine is automatically restarted and will generally show an error message (E02), due to the wrong checksum: this is normal and happens only when switching on the labeler for the first time, then the correct parameters checksum is automatically recalculated.

To help recovering operation on a machine locked due to improper configuration, it is also possible to activate this function at switch-on the Alpha Compact by pressing simultaneously the [Start] + [Stop] keys for at least 5 seconds.

PROGRAMMING RESET PARAMETER

- Press either [◀] or [▶] key* to move to the next parameter
- Press [Enter] to reset parameters (password request s. page **91**)
- To exit PROGRAMMING press the [Start] or [Stop] key

STORE PARAMETERS

This function allows customizing the default parameter set by changing the factory predefined values. Before executing this function, it is necessary to ensure that the current working parameters have been carefully tested to ensure proper and reliable machine operation (once executed, the original default values are overwritten). Being potentially dangerous, this function requires the 5-digit password (default 01234) to be executed, but it can also be totally disabled (refer to Service manual under chapter "Extended Configuration") by setting bit 1 of parameter 004.

Once executed, its correct execution is confirmed by a single "beep".

PROGRAMMING STORE PARAM.

- Press either [◀] or [▶] key* to move to the next parameter
- Press [Enter] to store parameters (password request s. page **91**)
- To exit PROGRAMMING press the [Start] (for labelling operation) or [Stop] (for stand-by operation) key

PASSWORD: 00000

If in programming a parameter is called up with [Enter] in order to change it, the machine asks for a PASSWORD to proceed, in order to limit access only to authorized personnel. The access code is "**01234**", once correctly entered it remains valid until the machine is switched off or reset by pressing [Stop] key two times.

The password is linked safely with the firmware by keyed parameters and may not be changed.

For avoiding confusion please note: In CONFIGURATION mode (s. page **90**) exists a 2nd inquiry of a password (with three digits).

```
PROGRAMMING        xxx
PASSWORD:        00000
```

- Press [Enter] to get access to the front number and/or press [Enter] again in order to jump to the rear numbers. Press either [◀] or [▶] key* in order to increase or reduce the value (corresponding cursor is flashing)
- Press [Enter] to confirm your input.

  With the valid password you will return to the configuration parameter.

  -With the invalid password appears "WRONG PASSWORD E15" in display.
- To exit settings, press the [Stop] key (for stand-by).

* Fast forward-function when pushing the button [◀] or [▶].

List of parameter PROGRAMMING

Safety instructions

HINWEIS

The PROGRAMMING MODE should only be edited by authorized and trained personnel!

- Wrong parameters can lead to bugs and improper functions, also possibly leading to mechanical damage.

The PROGRAMMING mode allows the consideration of all 256 service parameters and enables the access to a further sub-menu. For a change of the values, a 5-digit password has to be entered.

A detailed description of the programming mode, see the Service Manual Part No. 32708677.

Firmware History

The firmware is the ‘’operating system’’ of the Alpha Compact labeler’s control unit. Following is the chronological order of all firmware releases (including changes) until the publication of this Manual.

Version Alpha 01 Creation date: 27. December 2010
Prototype

Version Alpha 01a Creation date: 27. January 2011
Prototype

Version Alpha 01b Creation date: 29. March 2011
Prototype

Version Alpha 02 Creation date 04. April 2011
Prototype

Version Alpha 03 Creation date 27. May 2011
Prototype

Version Alpha 03a Creation date 11. June 2011

Version Alpha 03b Creation date 25. August 2011

Version Alpha 03c Creation date 17. November 2011

Version Alpha 04 Creation date 02. December 2011
9 configurations can be stored, first applicators are supported.

Version Alpha 04a Creation date 07. February 2012
Minor improvement of error diagnosis

Version Alpha 04b Creation date 14. February 2012
Minor improvement of error diagnosis

Version Alpha 04c Creation date 23. February 2012
Tamp-Blow applicator is supported

Version KD23 for Display Creation date June 2012
Display can be used for Alpha Compact and Legi–Air.

Hardware History

The hardware are the assembly parts of the Alpha Compact. Please find subsequently a chronological order of all hardware versions used until the publication of this Manual.

From 01.06.2012 (Serial number 1206xxxx)
New plastic sliding buch, belt pulley and label detection sensor are pluggable.

From 15.06.2012
Motor earthing connection is moved from lefthand bottom side to lefthand top side.

From 13.07.2012 (Serial number 12070050)
New unwinder arm

8. System Options

The labeler Alpha Compact can be equipped with many different options. The systems options are subdivided into three groups.

Additionally we offer many customized special solutions like protective cabinets, stands, special labeling stations and more. These options require a check and examination. Please contact therefore your System Consultant.

Subdivision of Alpha Compact Product Range

Pre-configured Alpha Compact (Basis version)

The labeler in Wipe On working method, for labels with a width of maximum 120mm or 150mm, in righthand- and lefthand version are the standard labelers. They are available ex-stock.

Retrofit Options (Field Installable)

Retrofit (Field installable) options are available ex-stock and can be installed at the Alpha Compact by the customer on-site with supplied installation instruction.

The most important retrofit options are:

- Manual in different languages
- Power cable for different power networks
- Pusher brushs
- Pusher rollers
- HMI Panel
- HMI Display
- Axial low label warning
- Alarm lamp
- Clamp box for i/O interface and product sensor input
- Pneumatic maintenance combination

Factory Installable Options (Factory Installable)

Options that are installed and tested at system order in our factory.

The most important factory installable options are:

- Language package 2 to HMI Display
- Shaft encoder
- USB-connection socket
- Zero Down Time Controller

Modular construction

The Alpha Compact is assembled based on modules according to special requirements of the customers in modular construction. There are assembled in our factory according to the required configuration.

The most important modules are:

- Alpha Compact Basis module
- Label detections
- Application with peeler blade
- Application with Tamp Blow Applicator
- Application with 90°/180° rotating tamp
- Unwinding

9. Maintenance

Safety Instructions

WARNING

Danger caused by direct or indirect contact to live parts

DANGER TO LIFE!

Contact of persons with live parts which are energized by failures.

– Disconnect the power from the machine before you perform any work at the labeler.

CAUTION

***[2]Danger caused by actively triggered movements.**

DANGER OF BEING CRUSHED!

Dependent on the application the products are transferred manually or automatically to the labeler.

– Perform maintenance work only if you have control of product supply and there is no danger.

CAUTION

Health Hazard due to highly combustible lubricants and detergents.

HEALTH HAZARD

Bin is under pressure and has inflammable content.

– Protect bin against sun radiation and temperatures higher than 50°C.
– Do not open after use.
– Do not sprinkle against flames or glowing items.
– Keep away from ignition sources.
– Do not smoke.

CAUTION

Health Hazard due to inappropriate handling of lubricants and detergents.

HEALTH HAZARD

For used lubricants and detergents, the valid information of the safety data sheets of the manufacturers and the valid safety- and disposal regulations have to be observed and followed for each product.

CAUTION

Danger caused by icing.

HEALTH HAZARD

Compressed air spray (21800768) is filled with liquid CO_2.

– Do not hold bin precipitate. Liquid CO_2 can escape of bin and icing occurs.

NOTICE

Damage to sensoric optic.

Sensoric devices have sensitive surface.

– Never use sharp-edged or hard objects for cleaning the feed-driver roller and sensors.

– Use only compressed air spray (21800768) for cleaning.

NOTICE

Damage to electric devices

Detergents can cause an electric short cut.

– Do not clean electric devices with water or liquid detergents.

– Wait after cleaning until detergents are dry.

NOTICE

Repair and maintenance work may only be carried out under switched off condition (disconnected from power) and may only be carried out by an electrician according to Accident Prevention and Insurance Association (UVV) 7.0 § 2 paragraph 3. Maintenance works that are only possible with power and/or air pressure switched on, may only be carried out by authorized and trained technicians who are familiar with existing dangers and regulations (e.g. AuS).

---

*2 Only if system is equipped accordingly.

Detergents

Use for cleaning our cleaning set, partno. 2180981.

In the cleaning set there is an instruction for cleaning, part-no. 22800512. This instruction informs about the necessary cleaning and gives advices for safe use of detergents.

Cleaning interval

Cleaning intervals depend on the environmental conditions and machine capacity (e. g. 3-shift work).

Refer to the recommended cleaning interval of the manufacturer.

In every case clean the labeler before changing labeling paper or ink ribbon.

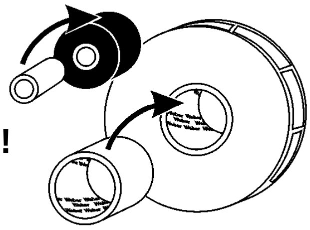
The often the labeler will be cleaned, the longer it will be ready for use and the quality of the labels is high.

REINIGUNGSHINWEISE / *CLEANING NOTES*

DE/GB

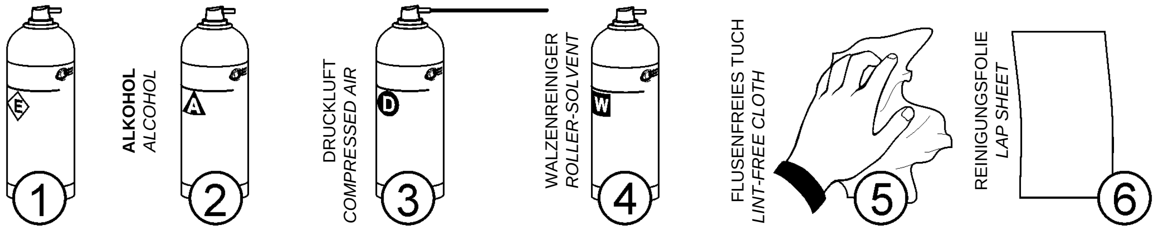
SERVICE-HOTLINE:
☎ +49 (0) 2224/ 7708 - 440
✉ hotline-ed@bluhmsysteme.com

REINIGUNG NACH JEDEM ROLLENWECHSEL !



CLEANING AFTER EACH ROLL CHANGE !

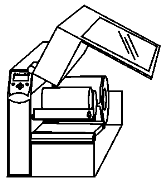
BESTELL-NUMMERN / *ORDER NUMBERS*



Nr.	Bestell-Nummer / *Order number*	Bezeichnung / *Description*
1	21800771	*LABEL REMOVER*
2	21800915	**ALKOHOL** / *ALCOHOL*
3	21800768	DRUCKLUFT / *COMPRESSED AIR*
4	21800977	WALZENREINIGER / *ROLLER-SOLVENT*
5	21800978	FLUSENFREIES TUCH / *LINT-FREE CLOTH*
6	21800921	REINIGUNGSFOLIE / *LAP SHEET*

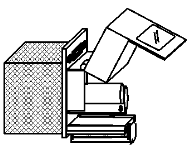
BESTELLANNAHME / *ORDER PROCESSING* ☎ **+49 (0)2224/ 7708-672** ✉ **bestellung@bluhmsysteme.com**

a



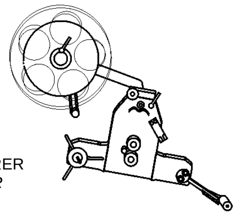
TISCHDRUCKER
DESKTOP PRINTER

b



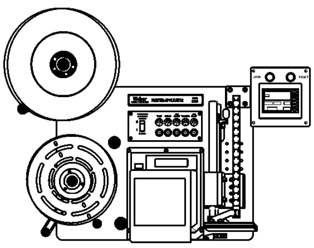
DRUCKMODULE
PRINT MODULES

c



ETIKETTIERER
LABELER

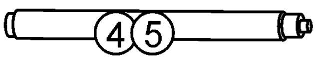
d



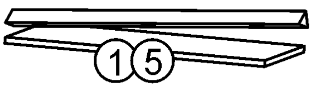
ETIKETTIERER
LABELER

a ✓ b ✓ c d — DRUCKKOPF / *PRINTHEAD*

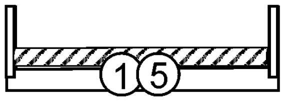
a ✓ b ✓ c d — GUMMIWALZEN / *PLATEN ROLLERS*



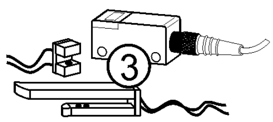
a ✓ b ✓ c ✓ d — SPENDEKANTEN / *PEELER BARS*



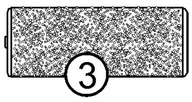
a b ✓ c d — FRIKTIONSEINHEIT / *FRICTION UNIT*



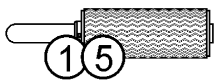
a ✓ b ✓ c ✓ d ✓ — SENSOREN/ *SENSORS*



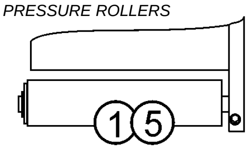
a b c ✓ d — ANTRIEBSWALZE / *DRIVER ROLLER*



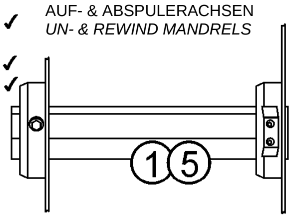
a b c ✓ d — FRIKTIONSWALZE / *FRICTION ROLLER*



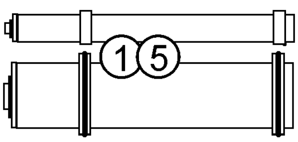
a b c ✓ d ✓ — ANDRÜCKWALZEN / *PRESSURE ROLLERS*



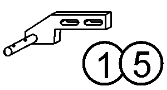
a ✓ b c ✓ d ✓ — AUF- & ABSPULERACHSEN / *UN- & REWIND MANDRELS*



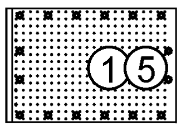
a b ✓ c ✓ d ✓ — FÜHRUNGSROLLEN/ *GUIDE ROLLERS*



a b c ✓ d ✓ — STÜTZLUFT-EINHEIT / *AIR ASSIST ASSEMBLY*



a b c ✓ d ✓ — STEMPELPLATTE /*TAMP PLATE*



Artikel-Nr./ Part-No.: 55203819 /18.08.2010

Daily Maintenance (After Approx. 8 Hours of Operation)

Requirements

- Labeler is free of energy (free of power and air pressure)
- No supply of products

Required Resources

- Roller solvent (*21800977)
- Alcohol (*21800915)
- Compressed air spray (*21800768)
- Lint-free cloth (*21800978)
- Label remover (*21800771)
- Soft brush (round or plain approx.. 10 mm)

* Product recommendation! Can be ordered at the Bluhm Weber Group by 8-digit article number.

Instruction

Please arrange the daily maintenance as follows:

Step	Procedure
1	Clean labeler according to the cleaning instructions.
2	*[2]Check / clean the air filter from the protective cabinet.

[2] Only if system is equipped accordingly.

Weekly Maintenance (After Approx. 40 Hours of Operation)

Requirements

- Labeler is free of energy (free of power and *²air pressure)
- No supply of products

Required Resources

- Roller solvent (*21800977)
- Alcohol (*21800915)
- Lint-free cloth (*21800978)
- Label remover (*21800771)
- Compressed air spray (*21800768)
- Soft brush (round or plain approx. 10 mm)

* Product recommendation! Can be ordered at the Bluhm Weber Group by 8-digit article number.

Instruction

Please arrange the weekly maintenance as follows.

Step	Procedure
1	Clean labeler according to the cleaning instructions.
2	*²Check / clean the air filter from the protective cabinet.
3	Clean all sensors (product sensor, low label sensor and label gap sensor) carefully with a soft brush or compressed air. .
4	Examine all roller assemblies for free rotation, if the axes of the rollers are fixed and correct, if necessary.

*² Only if system is equipped accordingly.

Six month Maintenance (After Approx. 1000 Hours of Operation)

Requirements

- Labeler is free of energy (free of power and *²air pressure)
- No supply of products

Required Resources

- Vacuum Cleaner

Instruction

Please arrange the six-month maintenance as follows.

Step	Procedure
1	Replace the *²air filter elements (if available).
2	Clean the system cover outside and *²the protective cabinet (if available) inside using an industrial vacuum cleaner.
NOTICE	Do not use air pressure from the shop air supply to blow dust from the machine. There is a possibility that this air can contain water and traces of oil, which can damage the electrical components in your machine!

Yearly Maintenance (After Approx. 2000 Hours of Operation)

Requirements

- Labeler is free of energy (free of power and *²air pressure)
- No supply of products

Instruction

Please arrange the yearly maintenance as follows.

Step	Procedure
1	Examine label liner rewinder and the friction clutch for excessive wear. Check all timing belts for wear and correct tension. Exchange all parts that are necessary for a correct function..
2	Check all moving parts for wear and bearing clearance.
3	Check all electric connectors and connecting devices.
4	Check all pneumatic connections.

*² Only if system is equipped accordingly.

.

Spare Parts

WARNING

Danger due to wrong spare parts!

HEALTH HAZARD

Wrong or faulty spare parts may affect the safety and may lead to injuries or damages at the system.

– Use only original spare parts or parts approved explicitly by the Bluhm Weber Group.

The spare parts of the labeler ALPHA COMPACT are included in a separate documentation (part number 32708614) and belong to the scope of supply.

10. Troubleshooting

Safety Instructions

WARNING

Danger caused by direct or indirect contact to live parts.

DANGER TO LIFE!
Contact of persons with live parts which are energized by failures.

– Disconnect the power from the machine before you perform any work at the labeler.

CAUTION

***2 Danger caused by actively triggered movements.**

DANGER OF BEING CRUSHED!
Dependent on the application the products are transferred manually or automatically to the labeler.

– Perform maintenance work only if you have control of product supply and there is no danger.

CAUTION

Health Hazard due to inappropriate handling of lubricants and detergents.

HEALTH HAZARD
For used lubricants and detergents, the valid information of the safety data sheets of the manufacturers and the valid safety- and disposal regulations have to be observed and followed for each product.

NOTICE

Check permanently all machine operations.

Abnormal noise development or movements (e.g. bucking, beating etc.) indicate failures and have to be checked.

Instructions for elimination of failures address only to trained personal.

Correcting Adjustments based on Labeler Result

Based on the labeler result you can draw conclusions from the necessary adjustments

The following requirements have to be complied:

- One or several sample products.
- Control of product supply.
- Labeler is connected with power supply and *²compressed air.
- Initial Operation is finished.

Product with a good Labeler Result

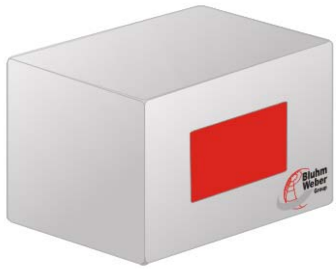


Fig. 10-1: Product with a good Labeler Result

Description:

The label is free from creases, straight and always at the same position on the product.

Instruction:

No corrections are necessary!

Product with creases in the Label

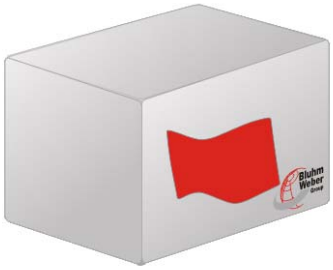


Fig. 10-2: Product with creases in the Label

Description:
The label has creases.

Instruction
Correct the labeler result as follows.

Step	Procedure
1	Examine the application speed.
2	The pressure of the pusher roller is too low. Adjust height of the pusher roller.
3	*²Examine the setup of rotating shaft encoder.

Product with beveled Label

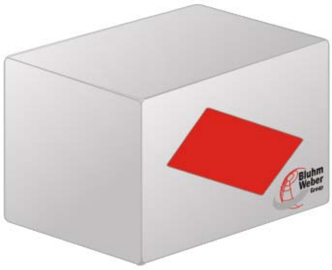


Fig. 10-3: Product with beveled Label

Description:
A label is beveled on the product.

Instruction
Please correct the labeler result as follows.

Step	Procedure
1	Correct inclination of the labeler and the peeler blade.

*²Only if system is equipped accordingly.

Product with Position Displacement of the Label

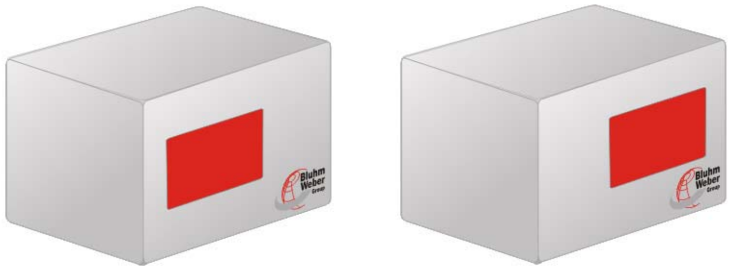


Fig. 10-4: Product with Position Displacement of the Label

Description:
The position of the label changes from product to product.

Instruction
Correct the labeler result as follows.

Step	Procedure
1	Check the product sensor.
2	Check distance from peeler plate to product.
3	Check the adjustments of the acceleration- and braking-ramp.
4	Check the stop sensor.
5	Check distance from guidance to the product supply.
6	Check the application speed.
7	*2Check the setup of rotating shaft encoder.

Label with incorrect position

Fig. 10-5:Label with incorrect position

Description:
The label is not in the desired position on the product.

Instruction

Correct the labeler result as follows.

Step	Procedure
1	Check the position of the product sensor.
2	Check the height setting of the peeler blade.
3	Check the product guiding of the conveying system regarding correct position and correct distance (guideline: approx. 1-2 mm)

Error Description

PROBLEM	POSSIBLE CAUSE	SOLUTION
Label liner breaks	• Damage of label roll • Nicks or label-cutter die damage on liner • Dents/damages at the side of the label roll • Liner width varies significantly • Peeler blade incorrect adjusted or damaged.	Exchange label roll.
	Adhesive residues at the area of the peeler bar.	Remove adhesive residues and check the label roll for damages caused by adhesive residues. Otherwise exchange label roll.
Label positioning on the product is incorrect.	The product is not constantly in correct labeling position.	Check if the conveyor`s speed is constant. Use guide bars or something similar to assure a constant product supply. If necessary, use a shaft encoder or tacho.
	Sensitivity of product-sensor is misaligned (if adjustable).	Check adjustment of the product-sensor
	Product-sensor and/or reflector are loose or vibrate.	
	Adjustment or position of sensor is in limit range.	
	Product-speed alternates.	Check and correct.
	Bad adhesive quality.	Exchange label roll.
	Object has a reflective surface, so that the start-light barrier reacts to reflections and not to the product edge.	Adjust light barrier a bit diagonal or rather use another light barrier. Maybe blind the lense: Therefore paste transparent strips over the lense.

PROBLEM	POSSIBLE CAUSE	SOLUTION
Multiple labeling of the product.	Start- or rather product-trigger-signal is bouncing (several impulses follow the trigger-signal).	Sensor is defect and has to be exchanged.
		In case of PLC-triggering check the outgoing signal and debounce if necessary.
The label is not applied (although the product passed the start-light barri-er).	No product detection.	Please clean reflector or sensor if they are polluted. Check adjustment of product-sensor (cp. manual of your sen-sor) or exchange it.
		The type of light barrier does not suit for the product because it`s transparent or reflective…
The desired rate of appli-cation cannot be reached with the labeler.	The required rate of appli-cation from the conveyor exceeds the specifications.	Check the system data consider-ing the used label size. Reduce product speed if necessary.
	Incorrect configuration pa-rameter values of the la-beler.	Check setup of the labeler.
	Slip at shaft encoder.	Check mechanical setup.
	[V-LABEL] is set too low also when using a shaft encoder.	Check the adjustment of speed-synchronization.
Labeler applies without evident reason.	Start-light barrier and/or product-sensor and/or re-flector are loose or vibrate.	Check adjustment of the sensor (cp. manual of your sensor).
	Start- /product sensor and reflector are not adjusted or misaligned.	
	Loose cable-connections.	Check all connections with the controller or inquire a Service Technician at the Service-Hotline

PROBLEM	POSSIBLE CAUSE	SOLUTION
Label does not stop constantly at peeler bar.	In case of label-stop-light barrier: Label is detected incorrect.	Load label material once again if necessary or check position of the label-sensor.
	Label-sensor ist polluted.	Clean label sensor.
	Band brake is adjusted too tight.	Check adjustment.
	In case of mechanical label-stop-scanning: Label is not correct detected.	Load label material once again if necessary or check position of the label-sensor.
		Remove buildup of adhesive at the scanning roll.
	Label is longer than 300mm.	Set the label size in the configuration menu in parameter [01Label length].
	Label sensor has switched to dark-light-recognation.	Press Teach button longer than 6 seconds. Sensor switches NO/NC).
	A non default pneumatic peeler blade is mounted.	Set distance between labelling sheet and label sensor and peeling edge [02Label position].
Labels run without a stop or rather the positioning of the label at the peeler bar is irregular.	Transmitter and receiver of the label stop-light barrier are not placed directly on top of each other (mechanical damaged).	Adjust transmitter and receiver and exchange if necessary.
	LED at receiver of the label stop-light barrier flashes permanently	You have to check the sensitivity-adjustment
Double-Labeling: On one trigger 2 labels are applied (directly) one after another.	The feed-length for an applying cycle is longer than two label lengths.	The overrun distance has to be reduced If this is impossible the driving ramps have to be shortened.
Labeler stops immediately.	At incorrect stress peaks, the labeler stops. (A possible hint for stress peaks is the jittering of the display controller).	Restart the labeler as usual. If the labeler stops increasingly often, remedy the causes of the stress peaks or install a mains filter before the labeler.

Error Messages via Display

As soon as an error was detected at the labeler, the ALPHA COMPACT stops and both relay signals (READY and SYNC) fall (passive). In the second line of the display an error message is indicated. Its meaning is described as follows:

NOT RESPONDING

Reset error after troubleshooting by pushing any button. The labeler changes automatically to ready mode and the READY-relay is active again.

DISPLAY MESSAGE	DESCRIPTION	CAUSE/ SOLUTION
NOT RESPONDING	This message indicates internal communication problems between controller and motor control.	Possible cause can be loose cable connections (inside and outside of the controller) or strong magnetic fields in the environment. If necessary, please let check by qualified personnel…
PARAMETERS ERROR	The memory-test when switching on the controller resulted in a checksum error.	Activated parameters might be damaged. If the error is still indicated after a new memory test, a chip in the controller is probably damaged. Please contact the Service Hotline (see page **9**).
TRIGGER ERROR	The trigger for application comes during application cycle.	Increase the distance between the products or increase the acceleration ramps.
L. EDGE NOT FOUND	The label sensor does not detect label material.	Check the setting of the sensor for label detection. Check whether there are labels on the label liner.

Additional Information on Signal Action in the Labeler

If the input signal of the product sensor changes from low (passive) to high (active), the following steps will proceed one after another:

1) Waiting for the adjusted time. (TRIGGER DELAY) ;
2) Registration of conveying speed via DIG (if connected and selected)
3) Activation of SYNC -relay
4) Label feed in predefined speed;
5) Waiting for the adjusted time SYNC PULSE;
6) Deactivation of SYNC -relay
7) Waiting for the adjusted time TRIGGER BLANK;
8) Label counter counts up;

If a failure is indicated after step 1, the action will be interrupted and therewith also the sequence.

While the labeler is waiting for the input trigger of the product sensor, the board will be controlled constantly, if a button is pushed.

If only the [**Enter**] button is pushed, it will equate to a manual triggering which will lead to the same sequence as already described.

Notes regarding Requirements to Label Material

The following requirements should be met for a correct operation of the standard label/stop-light barrier:

- Label material is not transparent.
- Label material has gaps of minimum 3 mm.
- Labels are stamped completely (without disposal fence or rather all labels are existing).
- Label liner is light-transmissive (opacity-check with light-barrier is possible).

Remarks regarding stress peaks

At heterodyned recurring transients (stress peaks in power network) the labeler is able to react to them and stop. If you do not deflect from standard software setting, the labeler returns to home position (like at turning on the labeler). Stored parameter are not deleted (minimum operating quality is maintained). To make the labeler ready for operation again, push the [Start] button. A possible remark for stress peaks is the jittering of the display controller. If the labeler stops increasingly often, it is advisable to remedy the causes of the stress peaks or install a mains filter or an undervoltage power supply before the labeler.

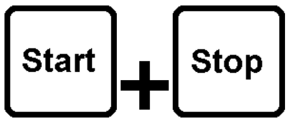
Set the labeler back to delivery status.

The parameter set that is entered when the system is delivered, is continuously stored in the controller. If parameter should be adjusted by mistake, there is the possibility to re-establish the delivery status.

Following requirements have to be fulfilled:

- Control about product conveyance.
- Labeler is connected to power.
- Start-up is completed.

Please reset the labeler to delivery status.

Step	Operating elements	Description
1		Disconnect the labeler from power supply.
2		Push the start + stop buttons at display controller or operating panel and keep the buttons pushed.
3		Connect the labeler with the power network.
4		Keep the start and stop buttons still 5 seconds pressed until the labeler is again ready for operation.

11. Index

12. EC-Declaration of Conformity

(DE) **EG-KONFORMITÄTSERKLÄRUNG**
(GB) EC-DECLARATION OF CONFORMITY
(FR) DECLARATION DE CONFORMITE CE

Weber Marking Systems GmbH
Maarweg 33
D-53619 Rheinbreitbach

DE **Wir erklären in alleiniger Verantwortung, dass die Maschine:**
GB We declare under our sole responsibility that the machine:
FR Nous déclarons sous notre responsabilité exclusive que la machine:

Alpha Compact

DE	Seriennummern:	von		bis	
GB	Serial number:	from	11080001	to	50129999
FR	Numéros de série :	du		au	

DE **auf das sich diese Erklärung bezieht, folgenden Bestimmungen und Richtlinien entspricht:**
GB to which this declaration relates corresponds to the regulations and directives.
FR que concerne cette déclaration, est conforme aux directives et réglementations suivantes:

2006/42/EG
2004/108/EG
2006/95/EG

DE **Bevollmächtigter für die Zusammenstellung der relevanten technischen Unterlagen:**
GB The person authorised to compile the relevant technical documentation:
FR Personne mandatée pour élaborer la documentation technique concernée:

Lutz Krämer
Maarweg 33
D-53619 Rheinbreitbach

Rheinbreitbach, 24. August 2011

Andreas Bluhm / Prokurist/ *Authorized Officer /* signataire autorisé